2D CAM FOR CNC · IN THE BROWSER

COPO

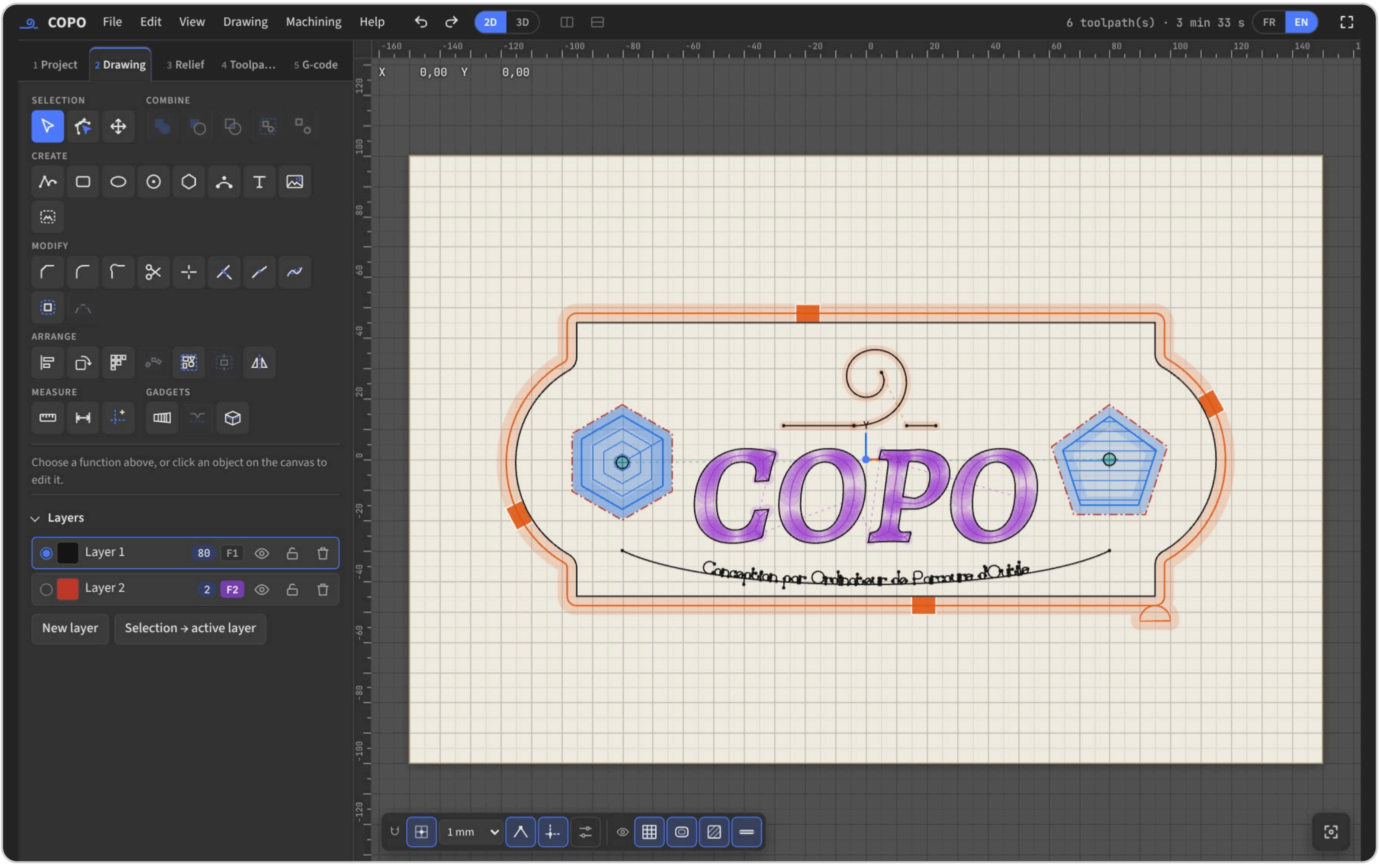
COPO draws your parts as vectors, calculates toolpaths (profile, pocket, V-carving, drilling) and writes the G-code for your machine. Everything runs in the browser, with no installation and no account.

`Version 1.253` `GPL v3` `Windows · macOS · Linux`

`Chrome · Edge · Firefox` `Français · English`

Open the app
https://gbulleryahen-hash.github.io/copo/

GitHub repository (code, download)
https://github.com/gbulleryahen-hash/copo


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
1 Project
2 Drawing
3 Relief
4 Toolpa...
5 G-code
SELECTION
COMBINE
CREATE
MODIFY
ARRANGE
MEASURE
GADGETS
Layers
Layer 1
80
F1
Layer 2
2
F2
New layer
Selection → active layer
FR
EN
X 0,00 Y 0,00
1 mm
COPO
Conception par Ordinateur de Parcours d'Outils

6 toolpath(s) · 3 min 33 s

Choose a function above, or click an object on the canvas to edit it.

The sample project: COPO text V-carved, a line of single-line text on a curve, a decorative flourish with a ball nose bit, two drilled holes, a cut-out with tabs and two pockets on side 2. Toolpaths are shown in color on the stock.

At a glance

Vector drawing

Rectangle, ellipse, polygon, star, arc, Bézier pen, text, nodes, chamfers, fillets, booleans, layers, dimensions.

2D machining

Outside, inside or on-the-line profile, spiral or zigzag pocket, rest machining, drilling.

V-carving

Along the medial axis, like Aspire: the bit rises up to the tip of every corner.

Tool entries

Back-and-forth ramp, continuous ramp, arc, ramped arc, helix. Allowance and finishing pass.

Tabs

Spaced automatically or placed by hand, flat or triangular.

Nesting

Places several parts on the sheet following their true shape, with rotations.

Single-line fonts

About sixty included (Hershey, EMS, Relief…) for fast one-pass engraving.

Double-sided

Layers per side, flip around X or Y, pause to turn the part over.

3D preview

Material removal simulation, side 1 and side 2, with animated playback.

G-code

GRBL, grblHAL, manual GRBL (G53), Carbide Motion, Onefinity, LinuxCNC, Mach3/4, UCCNC, MASSO, a custom post-processor, nearly sixty .pp-format machines (Haas, Axyz, Isel, Kuka, laser, HPGL…) and import of your own .pp files. G2/G3 arcs.

Setup sheet

Printable workshop sheet: 2D and 3D views, origin, tools, toolpaths, speeds, checks.

Millimetres or inches

Drawing, tools and feeds in mm or inches (fractions available), G-code in G21 or G20.

French and English

Interface and user guide in both languages, chosen with the FR / EN buttons at the top right.

Free software

GNU GPL v3 licence: use, modify and share COPO freely.

Your first job in six steps

① **Set up the project**

Project tab: stock size and material, X0 Y0 origin point, Z zero (top of stock or table), units and machine.

② **Draw or import**

Draw with the icons of the Drawing tab, import an SVG, DXF, PDF or AI file or an image (the **Import** icon in the Create group, or the File menu) or type some text.

③ **Check your tools**

Choose the material in the Project tab, then check in the **Tool library** (Toolpaths tab) the speed, feed, pass depth and stepover of each bit.

④ **Create the toolpaths**

Select shapes on the canvas, open the **Toolpaths** tab, choose Profile, Pocket, V-carve or Drilling, adjust, then click **Calculate**.

⑤ **Check in 3D**

The **Machining simulation** button (Toolpaths tab) or **Machining** › **3D machining preview** shows the machined stock. Play the job back to spot a wrong depth or tool.

⑥ **Save the G-code**

G-code tab: choose the post-processor, then **Save G-code**. Always do a dry run before cutting.

The interface

The window reads from left to right, in the order you work.

First launch: a welcome screen offers a **quick tour** (six bubbles on the main areas), the sample project or an empty project. Then, the first time you open each tab, one or more **tips** show the essentials. Skipping the tour also turns tips off. Use **Help › Quick tour** to see it again and **Help › Show tips again** to bring the tips back; the “Don’t show at startup” box hides the welcome screen.

- **Save button** (floppy disk, left of 2D/3D): the first time it asks where to save the project; after that it simply overwrites the same file (like `Ctrl` + `S`). A blue dot shows unsaved changes.
- **Menu bar**: File, Edit, Drawing, Machining, Extensions and Help. The Drawing menu sorts its commands into five framed categories: Groups, Combine, Vectors, Corners and Arrange. The **Extensions** menu gathers the modules that create drawings and toolpaths in one go: guitar fretboard, V inlay, pocket inlay, finger-joint box and rotary graduations. The Help menu opens this guide, lets you report a bug and shows the licence. The **FR / EN** buttons, at the far right of the bar, switch the interface language; the button next to them puts COPO in **full screen** (Esc to leave).
- **Drawing tab**: at the top, a grid of icons gathers every drawing function in two columns: on the left Selection and Library, Create, Arrange; on the right Measure, Modify, Combine; each with a tooltip and a shortcut letter. Just below appears what you need: the settings of the chosen tool (with a hint and a small diagram), the form of the clicked function (Align, Array copy, Nesting, Images…), or the Selection block when an object is selected. Greyed icons need a selection first. Layers stay at the bottom. Esc closes the open function.
- **Workspace**: the stock in beige, rulers in millimetres, the grid and the origin (X and Y arrows).
- **Left panel** with five tabs: **Project**, **Drawing**, **Relief**, **Toolpaths** and **G-code**. They are numbered 1 to 5 in the usual order of work. In the Project and Toolpaths tabs and in the tool library, small diagrams show what each setting does (depth, allowance, cut direction, tool entry, tabs, bit shape, Z zero…) and update with your choices.
- **Workspace bar**, floating at the bottom left of the 2D view (cursor coordinates relative to the origin show at the top left of the workspace, on a transparent background, like the 3D view information): two groups of on/off icons (their name shows on hover). *Snap* (magnet): grid and its spacing, objects, guides, and the settings button that opens Snap settings just above it. *Show* (eye): grid, toolpaths, removed material, overlapping lines. An active button has a

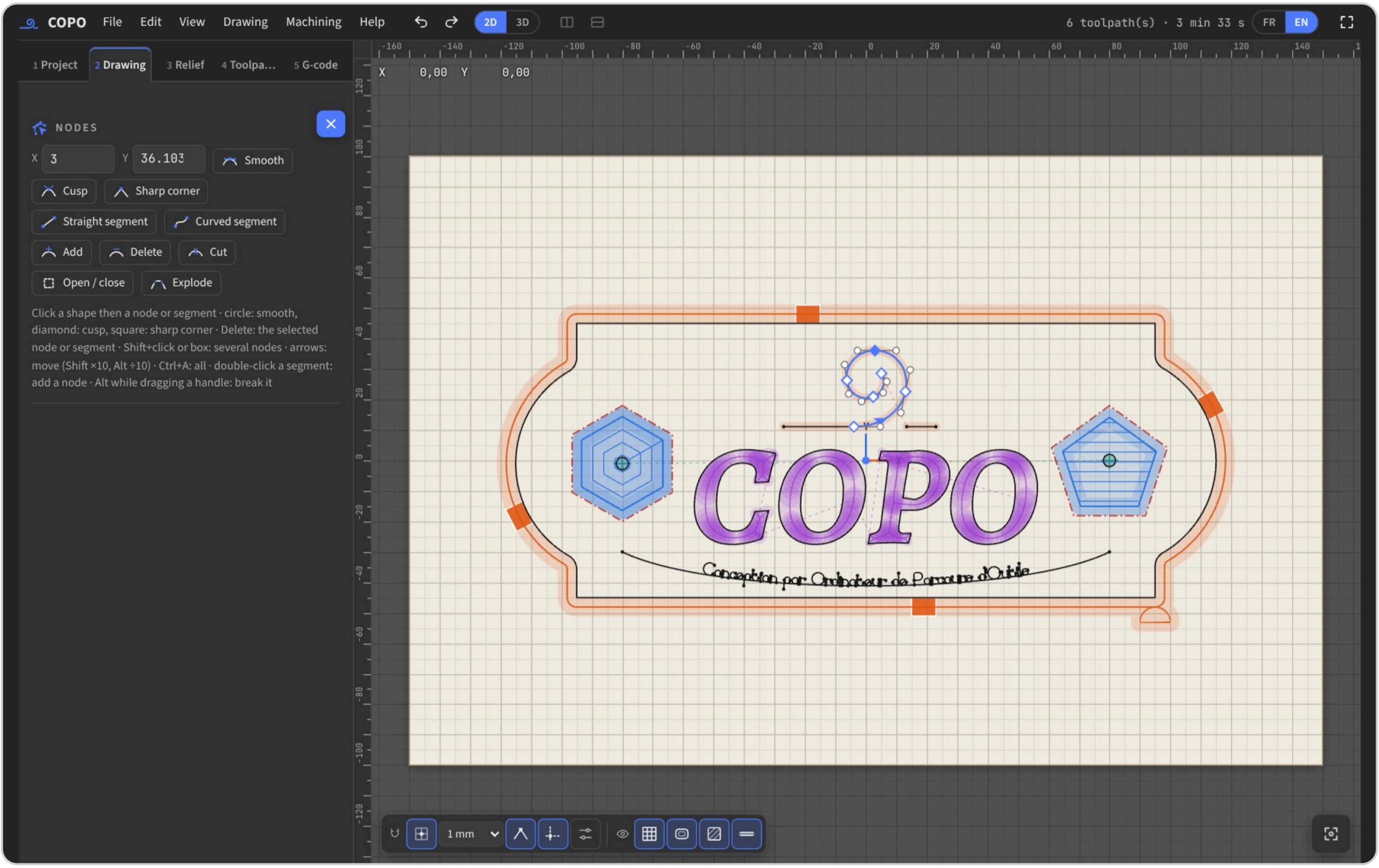
blue outline. At the bottom right, the **Recentre** button fits the view to the stock and drawing; the 3D view has the same bars in the same places.

- **Tooltips**: rest the mouse a quarter of a second on a button or setting to see what it does. The tooltip disappears as soon as the mouse leaves, and shows straight away on the next button while you explore. In the toolpath list only the name is shown: the operation, tool, time and any warnings (*!* badge) appear on hover.

A vector is selected by clicking **on its line** (not inside it), so stacked shapes stay reachable. An already selected shape can be moved by dragging from inside.

Right click (Select tool): on an object, in the 2D view as in the 3D view, it selects it and opens a menu next to the pointer. On a vector: **Machine** (Profile, Pocket, V-carve, Drilling), **Duplicate**, **Group**, **Weld**, **Offset…**, **Mirror…**, **Rotation…**, **Center on the stock**, **Create a relief** (Dome, Extrusion), **Move to layer** and **Delete**. On an image: vectorize, turn into a relief, fit, center, stacking order, lock, delete. On a relief: roughing, finishing or profile toolpath, outline, duplicate, fit, center, hide, delete, and in 3D move, rotate, resize, drop to the bottom or align to the top. On empty space on the 2D canvas, right click closes the current tool or settings and deselects, without a menu; on empty space in the 3D view: recenter, top view, front view. With the pen or another drawing tool, right click still ends the path; in 3D, right-dragging still pans the view. **Duplicate** (`Ctrl` + `D`) creates the copy *in place*, selected: drag it or move it with the arrow keys. **Overlapping lines** (duplicates, copies not moved, shared edges, doubled DXF imports) are highlighted in pink so you spot them at a glance: the cutter would pass twice over those portions. The *Overlapping lines* button in the Show group (or **Settings › Display**) turns this highlight on or off.

Drawing


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
6 toolpath(s) · 3 min 33 s
FR
EN
1 Project
2 Drawing
3 Relief
4 Toolpa…
5 G-code
NODES
Smooth
Cusp
Sharp corner
Straight segment
Curved segment
Add
Delete
Cut
Open / close
Explode
X 0,00 Y 0,00
COPO
Conception par Ordinateur de Parcours d'Outils
1 mm

X 3 Y 36.103

Click a shape then a node or segment · circle: smooth, diamond: cusp, square: sharp corner · Delete: the selected node or segment · Shift+click or box: several nodes · arrows: move (Shift ×10, Alt ÷10) · Ctrl+A: all · double-click a segment: add a node · Alt while dragging a handle: break it

The Nodes tool: X/Y position in absolute or relative mode, then the buttons sorted by use (node types, segments, add, delete, cut, explode, open / close).

The Drawing panel, as in Aspire: when you pick a tool or a function from the grid, its settings replace the icons. The **blue cross** at the top right, or a right click on empty space in the drawing, closes them and brings the grid back. When you click an object, the grid stays and its settings (Selection, image…) appear below it; help texts open with the ⓘ icon.

Shapes and paths

- **Rectangle** `R`, with square, external radius, internal radius (hollow arcs) or chamfered corners. **Ellipse** `E` (Shift: circle) and **Circle from center** `C`.
- **Typed values** (circle, ellipse, rectangle, polygon and star, arc): the tool panel shows fields as in Aspire. Circle: center X/Y, radius or diameter. Ellipse and rectangle: reference point (9-point grid), X/Y, width and height. Polygon and star: center, radius, angle. Arc: start point, end point, radius and height (sagitta), linked together. Draw with the mouse: the shape stays selected and the values change it directly (corners and number of sides included); **Confirm** finishes it, and you can draw the next one. With no shape in progress, type the values then **Create**. To come back later to a shape drawn this way (even

moved or resized), select it and click the pencil icon **Edit dimensions** in the Selection section.

- **Polygon and star** `O`: number of sides or points, inner radius. Click the centre then drag: the 2nd point snaps horizontally and vertically to the centre (`Ctrl`: 15° steps).
- **Pen** `P` (first tool of the Create row): click for a corner point, click-drag for a curve point, click the first point to close, double-click to finish. Near the horizontal or vertical of the previous point, the line aligns to it by itself (“Horizontal” / “Vertical” marker and dashes); with `Ctrl`, the angle moves in 15° steps (0°, 45°, 90°…). **Extending a line**: start the Pen on the end of an open line (“End” marker); the path continues that line instead of making a new one, and the curve to the next point leaves exactly along its axis (tangent). Drag from that first point, along the axis, to set the curve’s tension. The `space bar` ends the current line and starts a new one; a *right click* ends the line and leaves the pen. The **Next point** block in the panel creates points from the keyboard: type *X* and *Y* (absolute position), or an *angle* and a length *L*, or *dx* and *dy* from the last point. The six fields stay linked (typing the angle and length fills in X, Y, dx and dy) and a dashed preview shows the point. `Enter` or **Add** places it, **Undo** removes the last point, **Close** and **Finish** end the path.
- **Arc** `A`: through 3 points (start, end, then the bulge) or, in *Center, start, end* mode, click the center, the start then the end: the arc goes the way you turn the mouse around the center.
- **Nodes** `N`: move nodes and handles, double-click a segment to add a node, `Alt` while dragging a handle to break it, `Shift` to keep it horizontal or vertical to its node. Clicking a handle **selects** it (orange dot): its X/Y coordinates show in the panel, absolute or relative, and can be changed like a node’s. The *L* (handle length, its distance to the node) and *Angle* (0° to the right, 90° up) fields can be set too: changing L keeps the direction, changing the angle keeps the length. While dragging, nodes and handles snap to lines and to the **extension** of straight segments (dashed “Extension” marker): put a curve’s handle on the extension of the line before it and the curve is tangent to it. Four node types: **Smooth** (circle, aligned handles), **Symmetric** (dotted circle, aligned handles of the same length: dragging one mirrors the other), **Cusp** (diamond, two independent handles for a corner between two curves) and **Sharp corner** (square, no handles). Several nodes: `Shift`+click to add or remove one, drag a box in empty space to take them all at once (`Shift` to add), `Ctrl+A` for all. Selected nodes move together with the mouse or the **arrow keys** (1 mm, `Shift` 10 mm, `Ctrl` fine step depending on zoom, `Alt` ÷ 10). The X and Y fields give the node position in *Absolute* mode; in *Relative* mode, type an offset then Enter: it applies to all selected nodes. The buttons are sorted by use: node types (Cusp, Sharp corner, Smooth,

Symmetric), segments (curve, straight), add, delete, cut, then Explode and Open / close; Smooth, Sharp corner and Delete apply to all of them. A selected **segment** (click it, it turns red) is deleted with **Delete** or the `Del` key: a closed outline becomes an open line, an open line is split in two. To switch to another shape, click its outline. **Explode** (also in the Modify row) cuts the profile at every node: each segment becomes a separate vector.

Editing

- **Chamfer** `K` and **Fillet** `F` on one corner, or on every sharp corner from the **Corners** block.
- **Trim** `X` deletes the part of a vector between two intersections; **Cut** `U` splits a vector where you click.
- **Extend vectors** (Modify group): click the open line to extend, then move the mouse near one of its ends. Its extension shows as a dashed line, in the direction of the path, with a circle at each crossing with another vector. Click a crossing: the line is extended up to it. You can go on to the next crossing; `Esc` lets you pick another line.
- **Weld, Subtract, Intersect** for boolean operations; **Join** and **Close** for open vectors, gaps being bridged by a straight line or by a smooth curve that continues both vectors; **Offset contour**: on a closed vector outside, inside or both sides; on an open vector to the right or left of the path, or all around ("Both" creates a closed outline). Sharp or rounded corners, and an option to delete the original.
- **Smooth (fit curves)** (Modify): replaces vectors made of many small segments (tracing, DXF, import) with *arcs*, *Bézier curves* or *straight lines* that stay within the tolerance of the original. "Keep sharp corners" keeps as sharp corners those whose change of direction exceeds the max angle (60° by default). A dashed orange preview and the node count before/after are shown before applying; untick "Replace" to keep the originals. *Arcs* or *Bézier*? In COPO every curve (circles, arcs, ellipses, letters) is stored as Bézier curves, as in an SVG file. The *Arcs* mode finds the portions that follow a true circle (constant centre and radius) and draws them with Béziers fitted to that arc, at most one per quarter circle: the gap from a perfect arc is about 0.03% of the radius (under a hundredth of a millimetre for a 30 mm radius). Use it for mechanical shapes (templates, fillets, holes): the radii are truly constant and give clean G2/G3 arcs. The *Bézier curves* mode gives free-form curves, often with fewer nodes on organic shapes (traced drawings, scrolls), but a "round" part is not necessarily an exact circle.
- **Align and distribute**: choose the reference in "Align to". *The selection*: a single object aligns to the stock, several align to each other. *The first selected object*: it stays put and the others align to it; Distribute places them after it, in the

order you selected them, with the given spacing. *The stock*: aligns to the stock edges and distributes across its full width or height.

- **Rotation** (Arrange): an orange pivot appears at the centre of the selection; drag it on the plan (it snaps to objects, guides and the grid), pick one of the nine points, or type its coordinates. *Relative* rotation (the selection turns by the given angle, positive counter-clockwise) or *absolute* (the selection takes the given angle; COPO remembers earlier rotations). ↺ 90° and ↻ 90° buttons for quarter turns.
- **Handles**: click an already selected object a second time. Squares resize (corners keep proportions when “Keep proportions” is ticked, Shift for the opposite, Ctrl for whole millimetres; middles resize one axis), circles rotate around the pivot (Ctrl: 15° steps), and the pivot can be moved. Click again to hide them. The **Rotating frame** box (Selection block, unticked by default, under “Keep proportions”) makes the frame follow the angle of a rotated shape, so you stretch it along its own axes, as in Aspire (for a single shape, or shapes rotated by the same angle; a straight line takes its own direction, so its width is its length). The width and height of the Selection block are then those of the shape along its own axes, and changing them stretches it along those axes.
- **Moving with the keyboard**: the arrow keys move the selection by 1 mm (`Shift`: 10 mm). With `Ctrl`, the step follows the zoom: about one screen pixel, rounded to 1, 2 or 5 (0.05 mm close up, 1 mm from afar), to place an object precisely; `Ctrl`+`Shift`: ten times more. The step used is shown. Same for nodes and images.
- **Right-click and double-click**: a single right-click on the plan finishes the current line (polyline), cancels a shape in progress, goes back to the selection tool and deselects; with the selection tool, it hides the handles and then deselects. Double-clicking an object, from the selection tool or a shape tool (rectangle, ellipse, circle, polygon, arc, dimension, measure), selects it with its handles; another double-click opens node editing (Esc or right-click to leave).
- **Mirror** (Arrange group): flips the selection about the top, left, right or bottom edge of its frame, horizontally or vertically (about its center, or the stock center when the box is ticked), or about a straight line selected with the shapes (**Flip about the line**). Small icons (the letter P and its reflection) show the effect of each button. The *Create a mirror copy* box keeps the original in place.
- **Outline** (last icon of the Create group): with an image selected, opens its vectorization; with a relief selected, creates the vector that follows its edge.
- **Array copy** in a grid, a circle or *along a curve* (also the **Repeat along a curve** icon in Arrange): select the objects to repeat then, last, the guide curve (`Shift`+click); set the number of copies, spread over the whole curve or

with a spacing, a start offset, and tick *Align to the curve* so they rotate with it. **Mirrors**, **rotation**, **stacking order**.

- **Exact position and size**: **Selection** block, with a reference point chosen among nine (corners, middles, center).

Snapping, guides and dimensions

Edit › Settings › Snapping gathers the **snap settings**: snap to grid and its spacing, **guides**, stock corners, midpoints and centre, snap radius (small to large), and object snapping with its six kinds of points (ends and nodes, midpoints, intersections, centres, tangents, and *on path*: any point of a line or curve, the closest to the cursor). Precise points (end, midpoint…) take priority over "on path". **Centre of a circle or arc**: move the mouse near its outline and a small orange cross shows the centre; go to it to snap there (also for circles made of straight segments, imported or vectorized). The **origin** (X0 Y0) is always a snap point. When you move an object by grabbing one of its points (node, corner, midpoint, centre), that point is the one that snaps: drag a rectangle's corner onto the origin and it lands exactly there. These settings are remembered. Drag from a ruler to create a guide: drawn points snap to it, and when you move a selection its edges and centre line up with the guides (and with the stock if that box is ticked). **Double-click a guide** to open the **Guides** block (Measure group) on its position, to change it with the keyboard; the **padlock** on each row locks the guide, which no longer moves with the mouse. An orange symbol at the cursor shows the snap found. The grid icon below the canvas shows or hides the grid. The **Measure** tool `M` gives a distance and the **Dimension** tool `D` places a dimension that stays on the canvas. An end snapped to a node, a segment midpoint or the centre of an arc or circle (solid dot at the end of the extension line) stays attached to the object: move it, resize it or edit its nodes and the dimension follows, with its value updated. An end placed elsewhere (grid, intersection, free point) stays fixed.

Layers, images and files

- **Layers**: stroke color, visibility, lock and machined side (F1 top, F2 bottom).
- **Reference images** (**Import** icon › Image, **File** › **Insert an image…** or drag and drop): move, resize and rotate them with the handles to trace over them. They are never machined.
- **Drawing and image library** (**Library** icon of the Drawing panel, or **File** › **Drawing and image library…**): folders on the left, previews on the right. **Add to drawing** (or a double-click) places the chosen drawing (SVG, DXF, PDF, AI) or image (PNG, JPG…). **New folder**, **Add files…** and **Add a folder…** (a whole folder of the computer, with its sub-folders) fill it; **Import the current drawing** (also **File** › **Add the current drawing to the library…**) saves it as SVG, or just

the selection. **Export my library…** and **Import my library…** back it up or move it to another computer (.copolib file). The gear at the top opens **Settings › Library**: the library can stay in the browser, or be a **folder on the computer** (Chrome and Edge) made of real files, which you can also organise with the file explorer; preview size and double-click are set there too.

- **COPO library**: under “My library”, the **COPO library** folder holds 2D drawings (closed outlines, in mm) and grayscale reliefs supplied with COPO, sorted by category (2D drawings: corners, shapes, Christmas, panels and shields; Reliefs: animals, decorative, objects, plants, ribbons, textures…). They are free to use, even for parts you sell, and come from the website: you need to be online the first time, then they stay available offline. This folder is read-only: **Add to drawing** places the item, **Copy to my library** copies it into your own folders so you can rename or change it. A grayscale image is placed on the drawing; turn it into a relief in the Relief tab. The full list, with sources and licences, is also on the website’s Library page.
- **Adjusting an image** (Drawing panel, image selected): center, angle, width, height and opacity fit on two rows; the row of icons below fits the image to the stock, centers it, resets the angle to 0°, changes the stacking order, locks and removes it (each button's name shows on hover). Help texts open with the ⓘ icon.
- **Vectorize an image** (collapsible **Vectorize** section below the image settings; when collapsed it shows the threshold and smoothing): COPO traces the outlines of the image. Set the *threshold* (anything darker becomes the shape; “Auto threshold” works it out for you) and the *smoothing* (softer outlines, small details removed), tick *Invert* to outline light areas. The preview updates live; **Create outlines** adds closed, grouped shapes where the image is, ready for a pocket, a profile or V-carving. A sharp, high-contrast image (logo, line drawing, silhouette) gives the best result.
- **Import**: the **Import** icon in the Create group opens a panel with two choices, *Vector* (SVG, DXF, PDF, AI) and *Image* (template to trace over or vectorize).
- **SVG**: import in millimetres with curves kept; export the whole drawing or the selection.
- **PDF and AI**: the vector paths of the first page are imported to scale (in mm), curves kept; clipping areas, images and text that is not outlined are ignored. Illustrator files must have been saved with “Create PDF Compatible File” (the default); a very old AI (PostScript) is refused with a message. Convert text to outlines before exporting (Type › Create Outlines).
- **DXF** (text/ASCII, any version): lines, polylines with arcs, circles, arcs, ellipses, splines and blocks. The file's unit is respected (mm, cm, inches…); a file without a unit is read as millimetres. Touching segments are joined into outlines, closed when possible, and each DXF layer becomes a COPO layer.

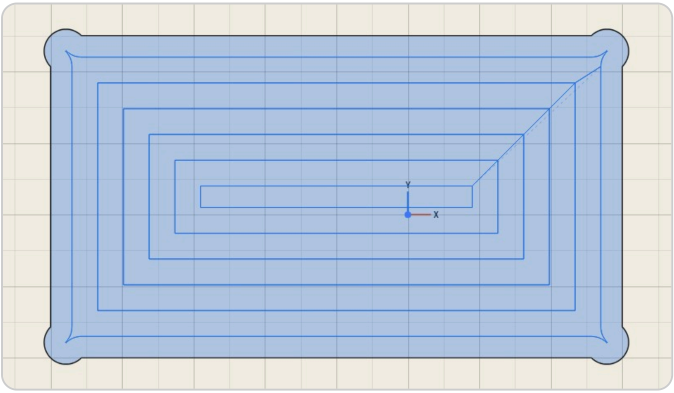
Text, dimensions, hatches and 3D objects are skipped (the message says which).

The drawing and image library: folders on the left, previews on the right.

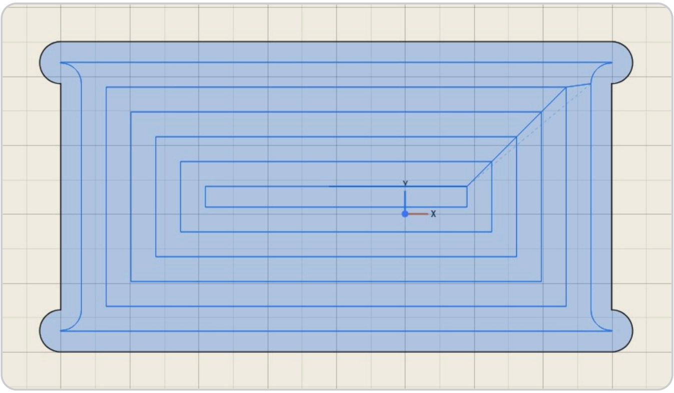
The drawing and image library: folders on the left, previews on the right, import and export buttons at the bottom.

Corners for tenons and mortises: dogbone and T-bone

A round bit leaves a fillet of its radius in corners, so a square tenon will not fit its mortise. The relief is drawn on the shape, corner by corner, so it only goes where it is needed. Take the **Relief** tool `B` (Modify group), choose *Dogbone* (the bit moves diagonally to touch the sharp corner) or *T-bone* (the relief extends the longer side, less visible) and the **bit diameter**, then click near the corners to relieve. The same settings are in the **Corners** block of the Drawing panel. Corners are picked one by one, as for chamfers and fillets. Relief only works between two straight segments; its diameter is 0.1 mm larger than the bit so it fits without forcing. Then machine the shape as usual (pocket, inside or outside profile) with that bit.

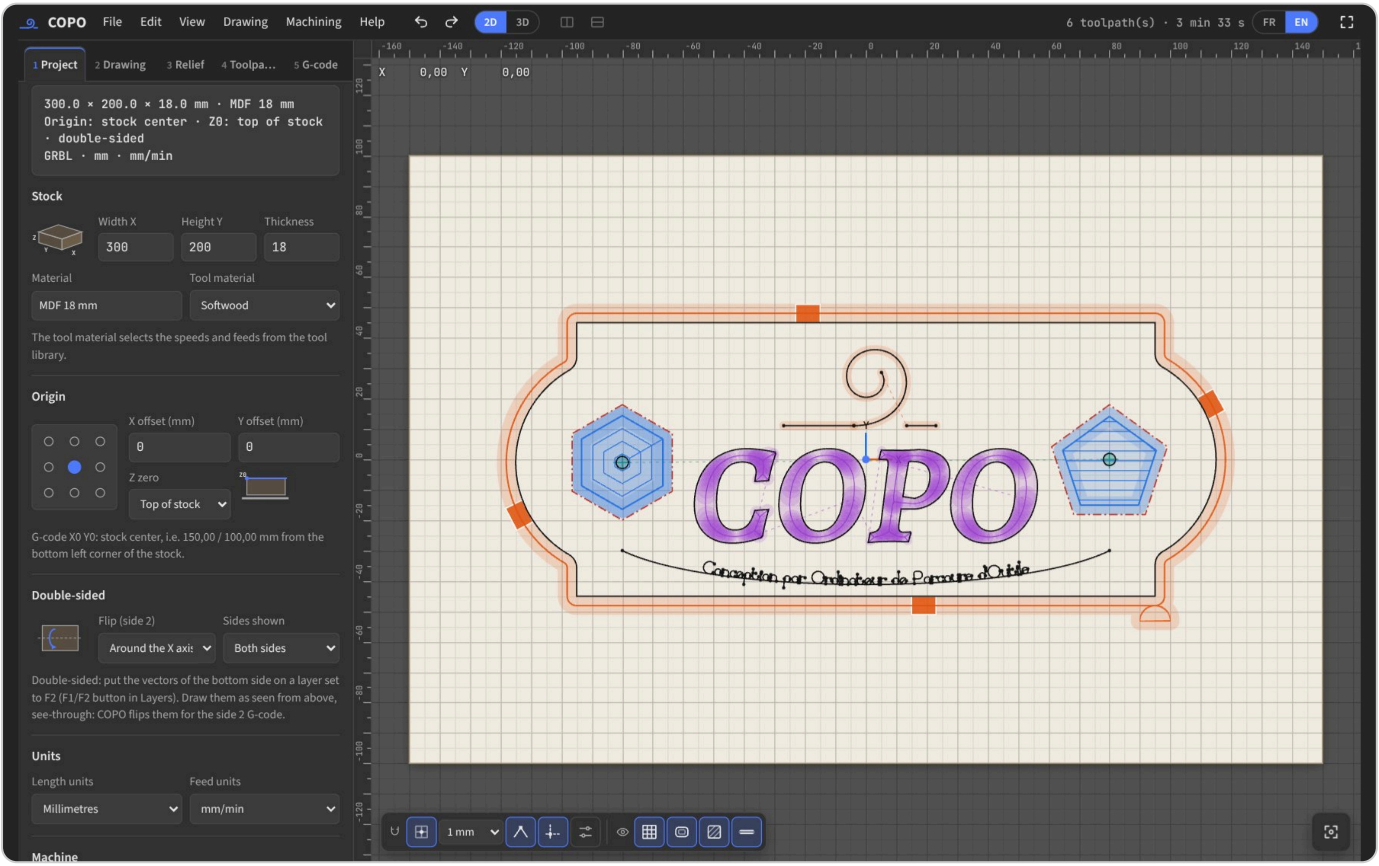


Dogbone: an arc the size of the bit, diagonally in each corner.



T-bone: the arc extends the longer side.

Project tab


COPO
File
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2D
3D
6 toolpath(s) · 3 min 33 s
FR
EN
1 Project
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3 Relief
4 Toolpa...
5 G-code
X 0,00 Y 0,00
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Conception par Ordinateur de Parcours d'Outils
1 mm

```
300.0 × 200.0 × 18.0 mm · MDF 18 mm
Origin: stock center · Z0: top of stock
· double-sided
GRBL · mm · mm/min
```

Stock

Width X: 300
Height Y: 200
Thickness: 18
Material: MDF 18 mm
Tool material: Softwood

The tool material selects the speeds and feeds from the tool library.

Origin

X offset (mm): 0
Y offset (mm): 0
Z zero: Top of stock

G-code X0 Y0: stock center, i.e. 150,00 / 100,00 mm from the bottom left corner of the stock.

Double-sided

Flip (side 2): Around the X axis
Sides shown: Both sides

Double-sided: put the vectors of the bottom side on a layer set to F2 (F1/F2 button in Layers). Draw them as seen from above, see-through: COPO flips them for the side 2 G-code.

Units

Length units: Millimetres
Feed units: mm/min

Machine

The Project tab gathers everything you set at the start of a job. The summary at the top lets you check it all at a glance.

The **Project** tab, first in the left panel, opens by itself when you create a new project. It contains:

- **Project type**: *Single-sided*, *Double-sided* (the Double-sided block appears) or *Rotary* (4th axis). The settings below change with the choice. If the selected machine has no rotary axis declared, a warning appears under the Rotary choice, without blocking anything.
- **Rotary**: the part is a cylinder turning between a chuck and a tailstock. *Job size*: length (L) and diameter (D). You draw on the **unrolled** surface of the cylinder, like a label laid flat: the sheet is L by π × D (the circumference, computed for you). The middle of the sheet is the top of the cylinder; dashed lines mark 90°, 180° and 270°, and the sheet edges are the seam (0° = 360°): a design crossing it must match up. *Z zero*: cylinder surface or axis. *Orientation*: rotation axis along X (Y wraps) or along Y (X wraps). The 3D view and the simulation show the part wrapped on the cylinder (untick to go back to flat, for example to place a relief with the handles). In the G-code, the wrapped coordinate is output as an **A** angle (or B, depending on the machine; degrees) with any post-processor, and arcs are replaced with short segments; a “Wrap” or “Rotary” post-processor (.pp) can also be used. Depths are measured towards the centre: the deeper you cut, the more the design tightens.
- **Stock**: X width, Y height, thickness and material (used by the setup sheet);
- **Origin**: the X0 Y0 point on the nine-point grid, an X/Y offset to place it outside the stock, and Z zero on the top of the stock or on the table;
- **Double-sided**: the flip axis and the sides shown;
- **Units**: millimetres or inches, feed unit;
- **Machine** (top of the tab): the name of the machine in use and its post-processor; **Change…** opens the **machine library**. On the left, the list of machines (the ticked one is used for the G-code; click another to switch), with **+ New machine**, **Duplicate** and **Delete** (two clicks to confirm). On the right, its settings, saved automatically: name, X/Y/Z travel (optional: the Project tab warns if the stock is larger), spindle speed range (tool speeds are clamped to it in the G-code), rotary axis fitted and its letter (A or B), notes. With the “GRBL manual” or custom post-processor, the *Rise to a machine Z (G53) at tool changes* box is **off by default**: a wrong machine Z can drive the spindle into its limit; when off, the spindle rises to the safe Z and G53 lines are removed, and: post-processor, rapid speed, **maximum cutting feed** (2000 mm/min by default: tool feeds are capped at this value), safe and clearance Z, G2/G3 arcs and comments. The **G81/G82/G83 drilling cycles** box writes holes as canned cycles: G81 plain, G82 with a dwell at the bottom, G83 with pecking, G98 retract to clearance Z. G83 has no dwell: a hole with both pecking *and* dwell is written as G0/G1. It is ticked by default for grblHAL, LinuxCNC, Mach3/4, UCCNC and MASSO, and unticked for GRBL, Carbide Motion, Onefinity and the custom post-processor (holes are then written as G0/G1); you can always

change it. If the controller does not know these cycles, it simply stops with an error.

The origin point must match the point where you set X0 Y0 on the machine. Stock, origin and material are saved with the project. Units and machine settings are remembered in the browser.

> Example: with the origin at the bottom left corner, a square whose corner sits at X0 Y0 on the canvas is cut from the bottom left corner of your sheet. The line under the grid tells you where the G-code X0 Y0 falls.

Rotary project (4th axis), with the sample

File › Open the rotary sample (4th axis) loads a beech cylinder, 90 mm in diameter and 150 mm long, with the word COPO V-carved and a flourish engraved all around. Follow it to see how it works:

Rotary sample: Project panel, unrolled sheet with angle marks and wrapped 3D view.

Left, the *Rotary* type and the job size; centre, the unrolled sheet (150 × 282.74 mm) with the 0°, 90° and 270° marks; right, the 3D view wrapped on the cylinder.

1. **Project type: Rotary**

   The stock becomes a cylinder. If the machine has no rotary axis declared, an orange warning appears, without blocking anything.

2. **Job size**

   Length 150 mm, diameter 90 mm: COPO computes the circumference (π × 90 = 282.74 mm), which becomes the width of the unrolled sheet.

3. **Z zero, XY origin, orientation**

   Z zero on the cylinder surface; origin middle left (X0 at the left end of the part, A0 on top); rotation axis along X, so Y wraps.

4. **Drawing and toolpaths**

   Draw and create toolpaths flat, as usual. The middle of the sheet is the top of the cylinder; its two edges meet (0° = 360°).

5. **Simulation and G-code**

   The 3D preview shows the material removed on the cylinder. In the G-code, Y becomes the A angle, in degrees, and arcs are replaced with short segments.

Simulation of the rotary sample: COPO carved on the cylinder.

Simulation at High resolution: the V-carved text is cut into the cylinder.

```
(COPO 2D CAM)
(Cylinder L 150 x D 90 mm unrolled sheet 150 x 282.743)
(XY origin: middle left - Z0: cylinder surface)
G21 G90 G17 G94
(Rotary axis: Y wraps to A - diameter 90 mm - Z0 at the surface)
G0 Z10
(Tool: Ball nose 3 mm)
…
(Rotary axis: Y wraps to A - diameter 90 mm - Z0 at the surface)
G0 X50 A14.137
X73.401 A14.14 F1000
X73.722 A14.148
X74.042 A14.162
X74.36 A14.181
```

Machine library

Machine library: list on the left, settings on the right.

Left, the machines (the ticked one is used for the G-code); right, the settings of the selected machine, saved automatically.

In the Project tab, **Machine** › **Change…** opens the library. Each machine keeps its post-processor, speeds, safe heights, X/Y/Z travel, spindle range and rotary axis (letter A or B). The default machine uses the *GRBL (M0 pause at tool change)* post-processor, and rising to a machine Z (G53) at tool changes is off by default. **Export…** saves all machines to a `machines-copo.json` file (another computer, backup); **Import…** adds the machines from such a file without replacing anything.

Settings

Edit › **Settings…** (`Ctrl` + `,`) opens a window with sections: titles on the left, content scrolling on the right (click a title to jump to it; the title of the part shown lights up). Everything is saved in the browser.

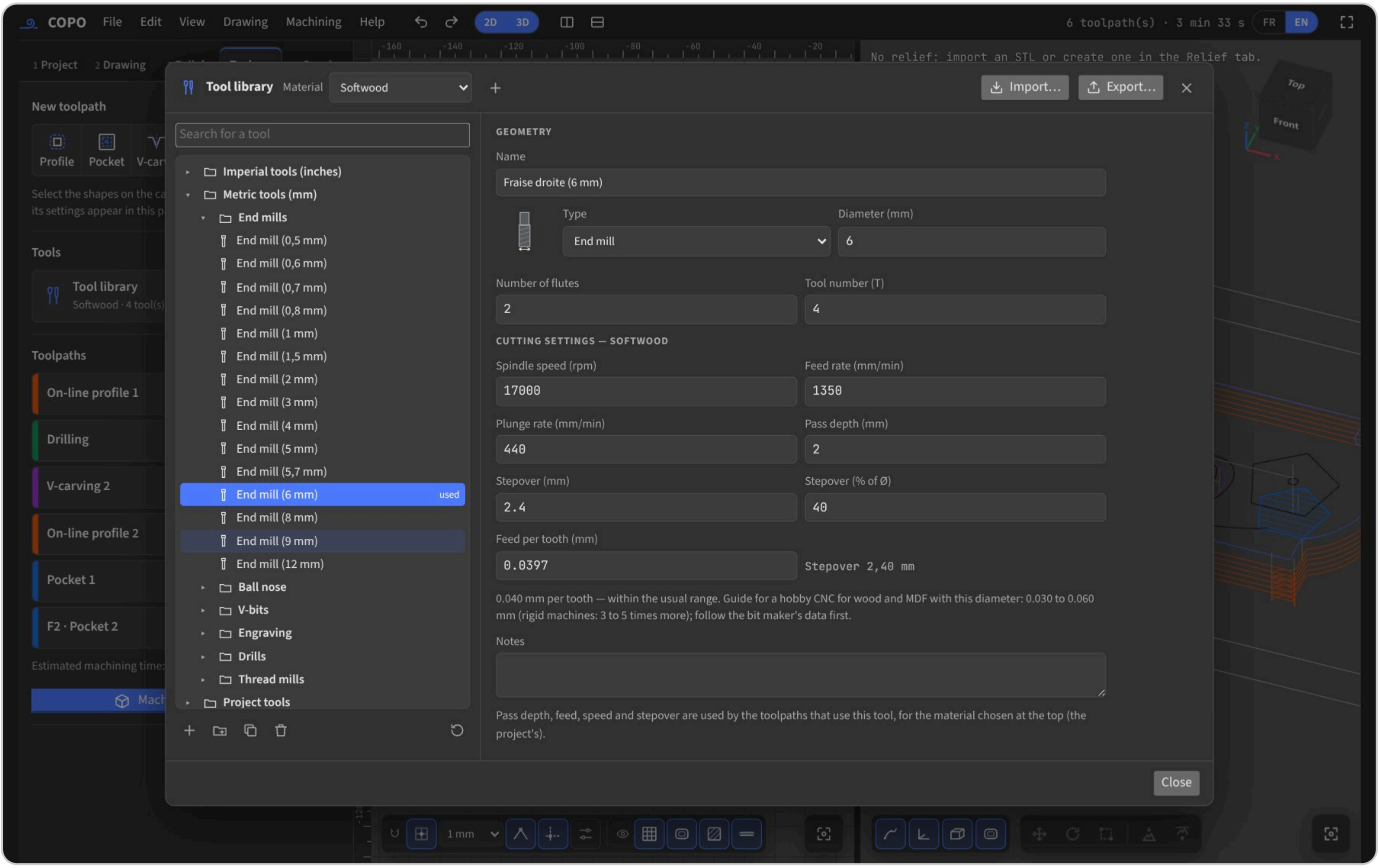
- **General**: interface language, welcome screen at startup, tab tips (and *Show all tips again*), quick tour.
- **Snapping**: snap to grid and its spacing, guides, stock, snap radius and object snapping (see Drawing). The snap button of the canvas bar opens this section

directly.

- **Display**: *Relief resolution*, as in Aspire: Standard (fastest), High (about 3 × slower, recommended) or Very high (about 7 × slower); whatever the resolution, a relief being moved follows the mouse immediately, shown slightly coarser, and the fine display returns on release; vertical walls (STL parts, relief edges) are smoothed, with no streaks or stair-stepped edges; toolpaths are not affected, grid, toolpaths, removed material, overlapping lines, ends of open lines, reliefs on the canvas.
- **3D views**: mouse buttons (by default like Fusion, FreeCAD, Tinkercad: hold right button = orbit, hold middle button = pan; or Classic: left button = orbit, right button = pan; Shift + right button pans on a touchpad) and orbit centre: Origin, View centre or Under cursor (default).
- **3D views** (continued): wheel zoom towards the cursor (default) or towards the view centre. **Relief colour**: grey (default), orange or blue, the same in the 3D view and in the simulation.
- **Library**: location of the drawing and image library (in the browser or in a folder on the computer), preview size, double-click action.
- **Simulation**: default resolution of the 3D preview (Standard, High or Very high).

Units and machine stay in the Project tab, close at hand. The G-code tab shows the selected post-processor; **Change…** opens the machine library.

Tool library


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
6 toolpath(s) · 3 min 33 s
FR
EN
1 Project
2 Drawing
New toolpath
Profile
Pocket
Tools
Tool library
Softwood · 4 tool(s)
Toolpaths
On-line profile 1
Drilling
V-carving 2
On-line profile 2
Pocket 1
F2 · Pocket 2
Estimated machining time
No relief: import an STL or create one in the Relief tab.
Tool library
Material
Softwood
+
Import…
Export…
Search for a tool
Imperial tools (inches)
Metric tools (mm)
End mills
End mill (0,5 mm)
End mill (0,6 mm)
End mill (0,7 mm)
End mill (0,8 mm)
End mill (1 mm)
End mill (1,5 mm)
End mill (2 mm)
End mill (3 mm)
End mill (4 mm)
End mill (5 mm)
End mill (5,7 mm)
End mill (6 mm)
used
End mill (8 mm)
End mill (9 mm)
End mill (12 mm)
Ball nose
V-bits
Engraving
Drills
Thread mills
Project tools
Close
Top
Front
1 mm

GEOMETRY

Name: Fraise droite (6 mm)

Type: End mill

Diameter (mm): 6

Number of flutes: 2

Tool number (T): 4

CUTTING SETTINGS — SOFTWOOD

Spindle speed (rpm): 17000

Feed rate (mm/min): 1350

Plunge rate (mm/min): 440

Pass depth (mm): 2

Stepover (mm): 2.4

Stepover (% of Ø): 40

Feed per tooth (mm): 0.0397

Stepover 2,40 mm

0.040 mm per tooth — within the usual range. Guide for a hobby CNC for wood and MDF with this diameter: 0.030 to 0.060 mm (rigid machines: 3 to 5 times more); follow the bit maker's data first.

Notes

Pass depth, feed, speed and stepover are used by the toolpaths that use this tool, for the material chosen at the top (the project's).

The tool library: the material at the top, tools sorted by unit system and type on the left, the tool's settings for that material on the right.

The library opens from the **Tool library** card in the Tools block of the Toolpaths tab (it shows the project material), from **Machining › Tool library…**, and from the **Tool** button of a toolpath window. Inspired by Aspire's, it comes filled: metric (mm) and imperial (inch) tools, sorted into end mills, ball nose, V-bits, engraving bits and drills, with settings for seven materials: **softwood, hardwood, MDF, acrylic, PVC, aluminium and brass**. The first four come from an Aspire library tuned for a WorkBee-type machine; settings missing there (large bits), PVC (like acrylic), aluminium and brass are **estimated** for a hobby CNC, at the low end of the ranges recommended for light machines (Shapeoko, X-Carve, 3018, Stepcraft; cross-checked with Carbide 3D and Sienci): in aluminium about 0.015 mm per tooth at Ø 3 and 0.025 mm at Ø 6, brass 20% lower. Estimated values are flagged by an orange banner: do a test cut and correct them. These values are deliberately cautious; makers' charts, written for rigid machines, go 3 to 5 times further.

- **Material** (top): this is the project material, also set in the Project tab (*Tool material*). Changing it makes every project tool take that material's settings and recomputes the toolpaths. The **+** button adds a material.

- **Tree** (left): click a folder to open it, double-click to rename it; the search box filters by name; “used” marks the tools of the project's toolpaths. At the bottom: new tool, new folder, duplicate, delete and back to the default library (two clicks). The **Import…** and **Export…** buttons are at the top right.
- **Settings** (right): geometry (name, type, diameter, angle, engraving tip, number of flutes, T number, notes), then cutting settings for the chosen material. The name updates itself as long as you have not edited it.

SETTING	PURPOSE
Type	End mill, ball nose, V-bit (V angle), tapered flat-tip engraving bit (angle and tip diameter), **tapered ball-nose** or drill (point angle, usually 118°).
Tapered ball-nose bit	Defined as on its box: **shank diameter D**, **tip radius R** and **cutting length L**. COPO derives the side angle, that of the cone tangent to the ball that reaches diameter D at height L: for D 6, R 1, L 20, about 5.99° (11.99° included). Engraving and tapered bits are used like V-bits of that angle (V-carving, on-the-line profile); the rounded or flat tip is not yet taken into account in the depth.
Diameter	Used to calculate offsets and pockets.
Tool number (T)	Written in the G-code of tool-change post-processors (`T2 M6`, `M6 T2`).
Spindle speed	In rpm, written with `M3`.
Feed, plunge rate	In mm/min, for cutting and for plunging or ramping.
Pass depth	Maximum depth per pass; it fills the toolpath's pass depth when you choose the tool.
Stepover	Distance between two pocket passes, in mm or % of the diameter (as in Aspire).
Number of flutes	Number of cutting edges of the bit.
Feed per tooth	Calculated: feed ÷ (spindle speed × number of flutes). Type a value to recalculate the feed. The comment below, on an orange background when conditions are not optimal, says whether it is low (the bit rubs and heats up) or high (risk of breakage), against the hobby-CNC ranges (wood and MDF, plastics, aluminium, brass) by diameter, and warns when the feed exceeds the machine's maximum feed; the bit maker's data come first. For a drill, COPO gives the feed per revolution.

Choosing a toolpath's tool

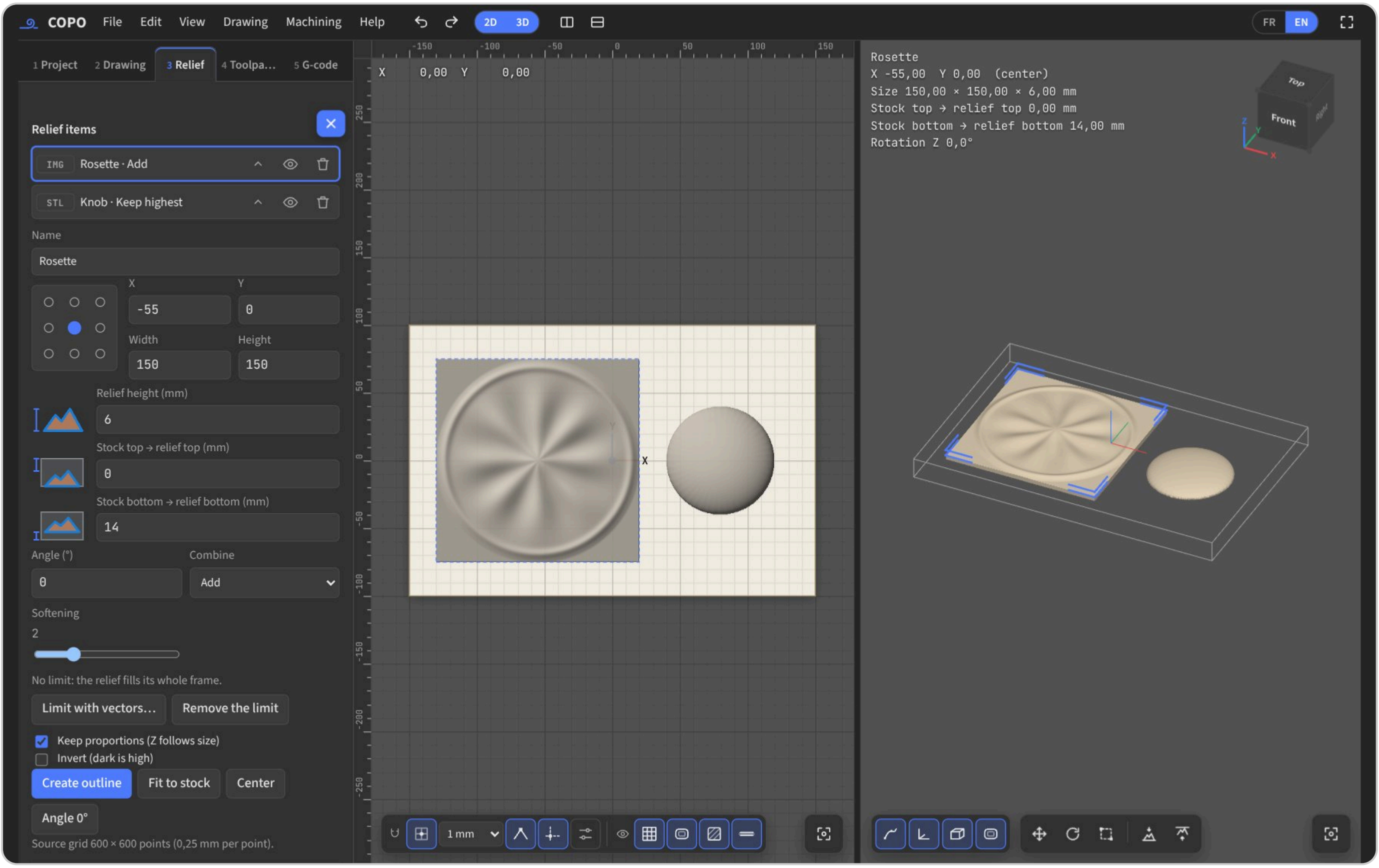
In a toolpath window, the **Tool** button shows the chosen tool; click it to open the library, select the tool then **Use this tool** (or double-click). Tools that do not suit the toolpath type are greyed out. The toolpath takes the tool's settings for the project material, and its pass depth. Editing a tool in the library updates the toolpaths that use it.

Importing an Aspire library

The import button reads Aspire, VCarve and Cut2D libraries directly (`.vtdb` file, for example `C:\ProgramData\Vectric\Aspire\V12.0\ToolDatabase\tools.vtdb`, or an export from Aspire's library). Tools, their folders and their per-material settings are added in an “Aspire import” folder, inches converted to millimetres; if the file holds several machines, COPO asks which one to use. Missing materials (PVC, aluminium, brass) are estimated. Form tools and diamond drag bits, which COPO cannot machine yet, are skipped. Import also accepts a COPO library exported as `.json`.

Relief (modelling)

The **Relief** tab prepares sculpted machining, like Aspire's modelling: each relief item is a height map placed in the stock. The tab opens the 3D view next to the drawing (**2D** / **3D** buttons); the creation buttons show how to use them on hover.


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
FR
EN
1 Project
2 Drawing
3 Relief
4 Toolpa...
5 G-code
Relief items
Limit with vectors…
Remove the limit
Create outline
Fit to stock
Center
Angle 0°
1 mm
Top
Front
Right

IMG Rosette · Add

STL Knob · Keep highest

Name: Rosette

X: -55
Y: 0
Width: 150
Height: 150
Relief height (mm): 6
Stock top → relief top (mm): 0
Stock bottom → relief bottom (mm): 14
Angle (°): 0
Combine: Add
Softening: 2

No limit: the relief fills its whole frame.

☑ Keep proportions (Z follows size)
☐ Invert (dark is high)

Source grid 600 × 600 points (0,25 mm per point).

X 0,00 Y 0,00

```
Rosette
X -55,00  Y 0,00  (center)
Size 150,00 × 150,00 × 6,00 mm
Stock top → relief top 0,00 mm
Stock bottom → relief bottom 14,00 mm
Rotation Z 0,0°
```

A grayscale image (rosette) and an STL model (knob, "keep highest") with the item settings.

- **Grayscale**: insert the image (PNG, JPG) in the Drawing tab (**Import** › **Image**), place and size it, select it, then click **Grayscale**: the image is *turned into a relief* at the same place, size and angle, and leaves the 2D view. The relief is shown shaded on the workspace and handled like an object. Light is high, dark is low (tick *Invert* for the opposite). A transparent background counts as black for heights but is not machined: with no boundary selected, machining stays within the opaque part of the image.
- **3D model** (binary or text STL): the model is seen from above and projected into heights, at its real size: it is never scaled down, even if it is larger than the stock. In that case, a message and the panel summary (in red) say so: enlarge the stock in the Project tab, or shrink the model yourself. If the model comes in lying down or upside down, *X angle* and *Y angle* tilt it (180° on X to flip it, ±90° to stand it up), *Z angle* turns it flat (other reliefs only have the Z angle).

Undercuts are machined as seen from above. Also from **File** › **Import an STL (relief)…**.

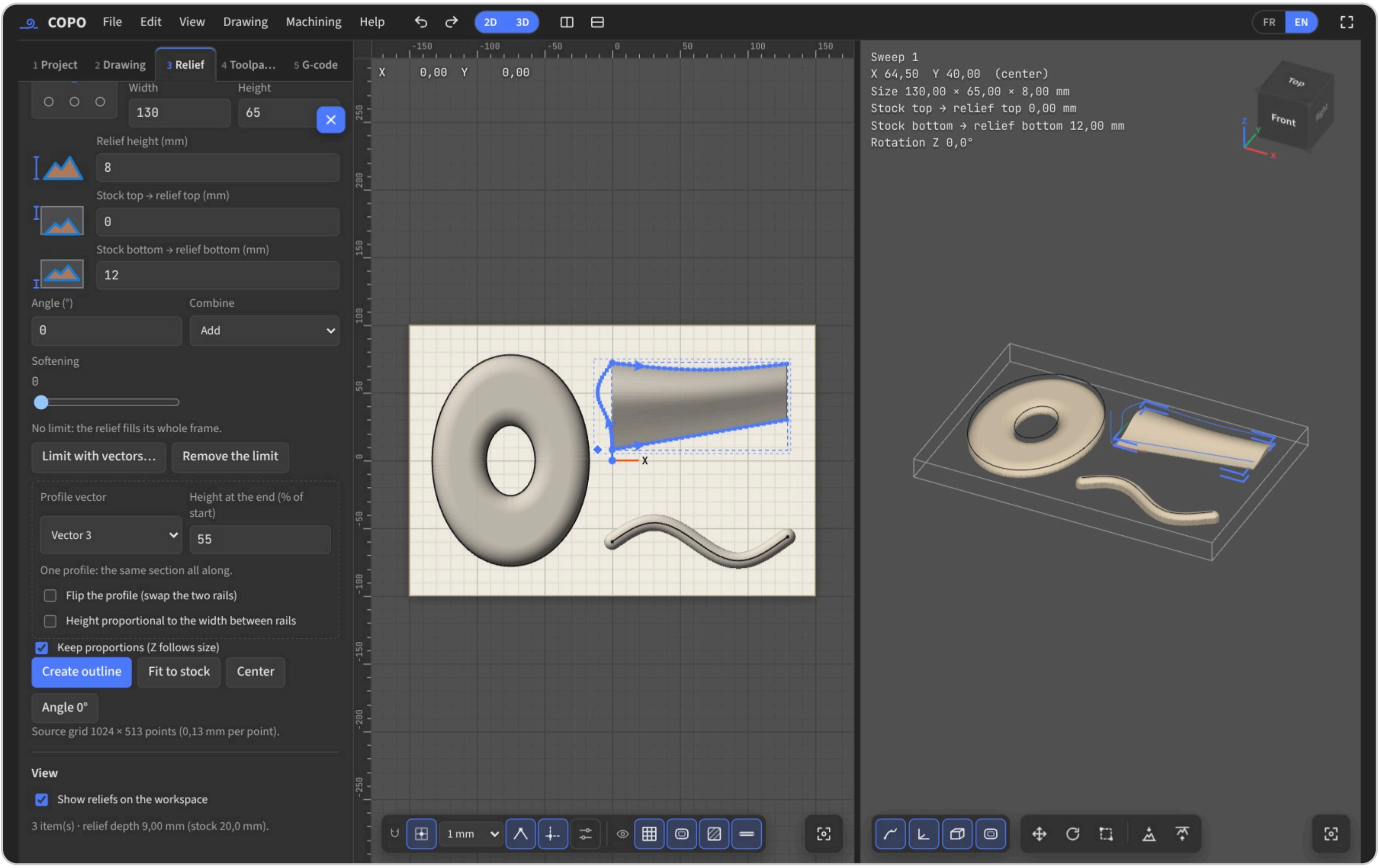
- **Settings of each item**: position and size as for an object (one of nine reference points, X, Y, width, height), *relief height*, *stock top → relief top* and *stock bottom → relief bottom* (editing one recalculates the other: their sum plus the height is the stock thickness), angle, *softening* and how it *combines* with the items before it: add, subtract, keep highest or keep lowest, or *blend*: where reliefs overlap, their heights are averaged, faded towards the item's edges, without adding or subtracting. "Keep proportions" makes the height follow size changes. To delete an item, click the bin on its row in the list. When a relief is selected, its settings take the place of the creation buttons; the blue cross, or a right click on empty space in the drawing, closes them.
- **Limit with vectors**: click **Limit with vectors…**, pick one or more closed vectors on the workspace and click **Apply the limit**. The relief only exists inside those vectors (a shape inside another makes a hole); outside there is no relief. The limit follows the vectors if you edit them; **Remove the limit** gives the whole frame back to the relief.
- **Create outline**: with a relief selected, this button creates a closed vector that follows its outline seen from above (STL, image, dome, sweep…), taking its rotation and limit into account. Then apply an outside *Profile* toolpath to cut the part out, or offset it with *Offset*.
- **Cutting out a relief without drawing its outline**: in a *Profile* toolpath, click the relief itself, on the 2D canvas or in the 3D view. Its outline shows as a dashed blue line and the button becomes **Compute around the relief**. The depth is set automatically to the bottom of the relief (stock top → relief top + relief thickness, never more than the stock thickness); you can change it. The toolpath follows the relief's outline and is recomputed when you move or change it (update it). Click the relief again to remove it; you can also add vectors to the same toolpath.
- **On the workspace**, a relief (STL, grayscale, dome, extrusion or sweep) is handled like an object with the Select tool: a first click selects it and opens its position settings in the Relief tab, and you can drag it to move it; a second click shows the handles: squares to resize (corners: proportions kept, Shift to distort; sides: one dimension only), circles to rotate (Ctrl: 15° steps); Ctrl resizes in whole millimetres. Delete removes it. Right-clicking a relief, on the canvas or in the 3D view, opens its menu (toolpaths, outline, duplicate, delete…).
- **Display**: in 2D, all reliefs (STL, grayscale, domes, extrusions and sweeps) are shown shaded under the drawing; the whole relief is visible in the 3D view. The summary gives the depth of the lowest point and warns if it exceeds the stock thickness. A relief that extends beyond the stock (after shrinking the

stock, for example) stays fully shown, in 2D and 3D, but only the part inside the stock will be machined.

Relief data is kept in the browser and saved in the `.copo` file: each relief's heights, images, and the full mesh of STL models. A project sent to someone therefore opens as is on another computer, without the original STL file, and the model's orientation can still be changed.

Shapes built from the drawing

Select vectors on the workspace, then click one of the three buttons under **Create from the drawing**. The shape is copied into the relief: editing the drawing afterwards does not change it.


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
FR
EN
1 Project
2 Drawing
3 Relief
4 Toolpa…
5 G-code
Limit with vectors…
Remove the limit
Create outline
Fit to stock
Center
Angle 0°
1 mm
Top
Front
X

Width: 130
Height: 65
Relief height (mm): 8
Stock top → relief top (mm): 0
Stock bottom → relief bottom (mm): 12
Angle (°): 0
Combine: Add
Softening: 0
No limit: the relief fills its whole frame.
Profile vector: Vector 3
Height at the end (% of start): 55
One profile: the same section all along.
☐ Flip the profile (swap the two rails)
☐ Height proportional to the width between rails
☑ Keep proportions (Z follows size)
Source grid 1024 × 513 points (0,13 mm per point).
View
☑ Show reliefs on the workspace
3 item(s) · relief depth 9,00 mm (stock 20,0 mm).

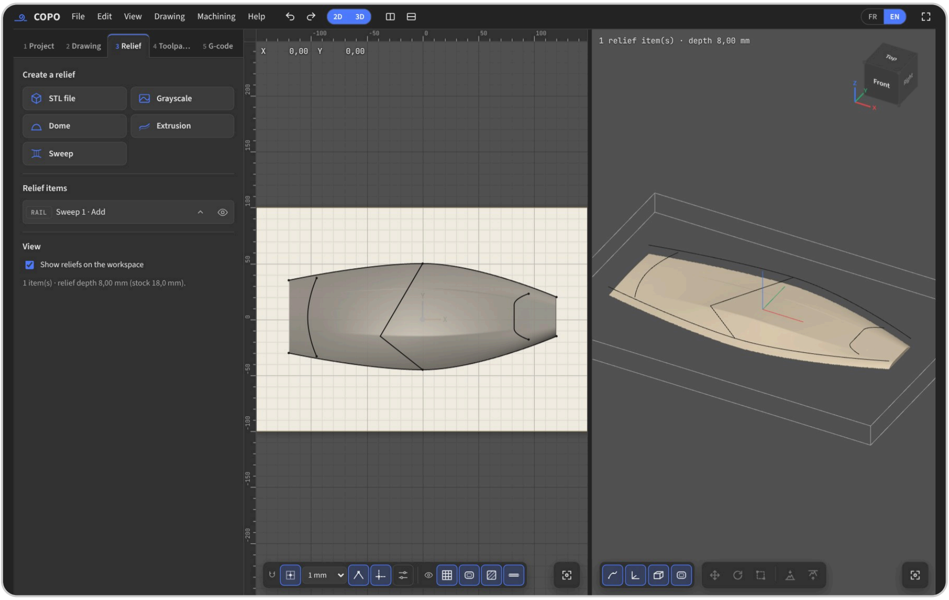
X 0,00 Y 0,00

```
Sweep 1
X 64,50  Y 40,00  (center)
Size 130,00 × 65,00 × 8,00 mm
Stock top → relief top 0,00 mm
Stock bottom → relief bottom 12,00 mm
Rotation Z 0,0°
```

A dome with a 14 mm edge on an ellipse with a hole, an extrusion along a curve and a two-rail sweep with an ogee profile (end height 55%).

- **Dome** (one or more closed shapes; inner shapes make holes): height rises from the edge. *Edge width* = 0 domes up to the center (a ridge along elongated shapes, as in Aspire); a value gives a rounded edge of that width, then a plateau.
- **Extrusion** (one or more vectors, open or closed): a moulding of the chosen *width* follows the path, with rounded ends.

- **Sweep** (two rails + one profile vector, all open): the drawn section spans from one rail to the other, scaled to the gap between them. The profile is the vector joining the two rails, otherwise the last one selected (changeable with *Profile vector*); its chord runs from rail to rail and its real height gives the Z height; *Flip the profile* reverses it if needed. Without a third vector, the section is round; *Height at the end* varies the height from start to end (scroll, neck, domed edge) and *Height proportional to the width* adapts it to the gap between the rails.
- **Sweep with several profiles**: to make the section change along the rails, draw several profiles *across*, each from one rail to the other, where it should apply (both ends on the rails). Select both rails and all the profiles, then **Sweep**: COPO recognises the rails (the vectors the profiles touch), places each profile where it crosses the rails and blends the section gradually from one to the next, shape and height included. Before the first profile and after the last, the section stays that of the nearest profile. The *Smooth transition between profiles* box (ticked by default) rounds the change from one section to the next: no kink at a profile, and the shape arrives flat at the first and last ones; untick it for a straight-line transition. The panel shows the number of profiles and their position as a % of the rails' length. To move a profile, move its vector on the workspace and create the sweep again.



X 0,00 Y 0,00

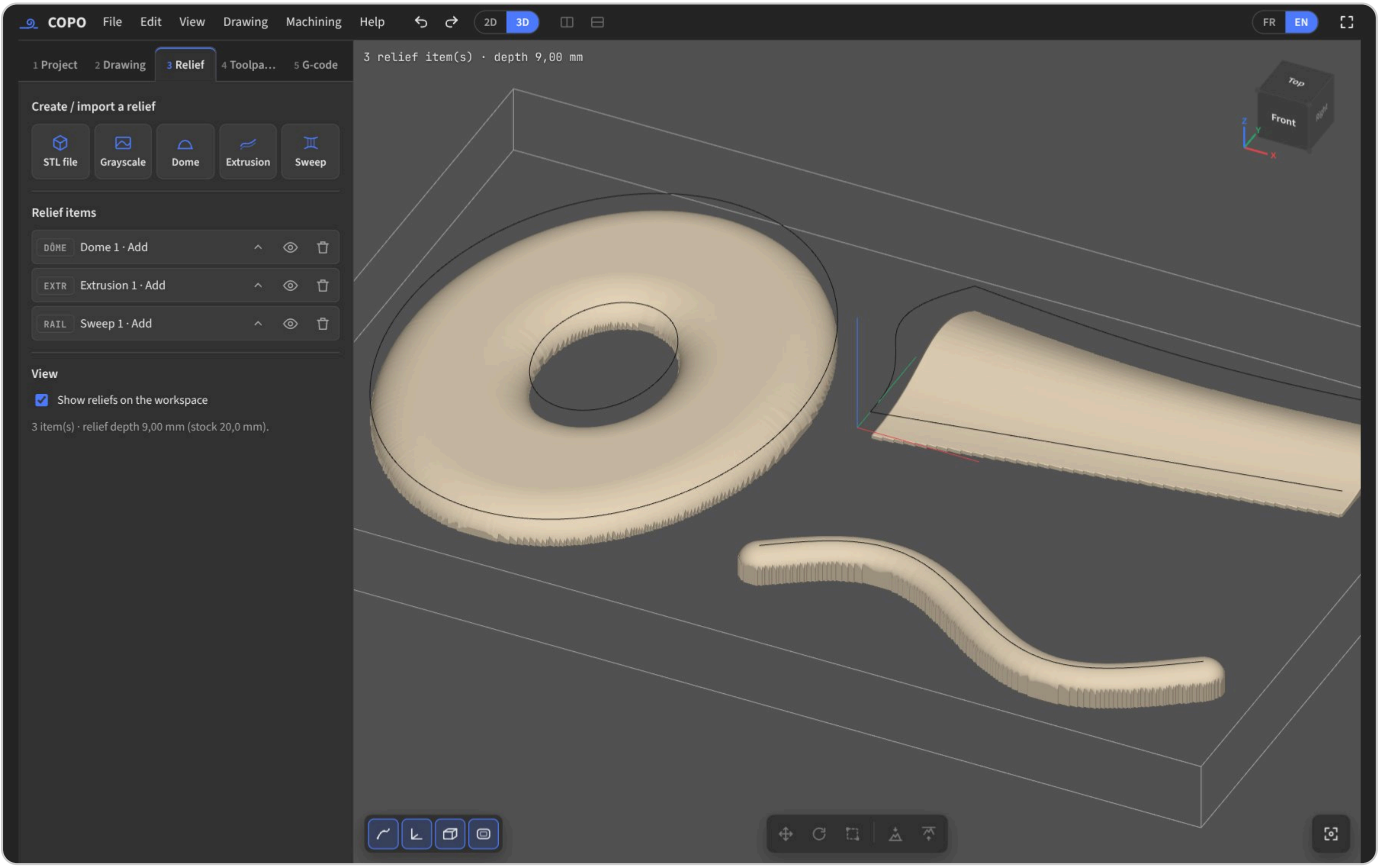
☑ Show reliefs on the workspace

1 item(s) · relief depth 8,00 mm (stock 18,0 mm).

1 relief item(s) · depth 8,00 mm

Two rails and three profiles drawn across: the section goes from a low round bump (left) to a high peak (middle) then to a flat section (right), with a smooth transition.

- **Profile** of domes and extrusions: round, angular (pointed), flat or ogee (S-shaped). Z height, base offset, combine mode and softening work as for other items.


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
FR
EN
1 Project
2 Drawing
3 Relief
4 Toolpa...
5 G-code
Create / import a relief
STL file
Grayscale
Dome
Extrusion
Sweep
Relief items
DÔME
EXTR
RAIL
View
Top
Front
Right

Dome 1 · Add

Extrusion 1 · Add

Sweep 1 · Add

☑ Show reliefs on the workspace

3 item(s) · relief depth 9,00 mm (stock 20,0 mm).

```
3 relief item(s) · depth 9,00 mm
```

View the relief in 3D: the relief carved into the stock, its highest point at the surface.

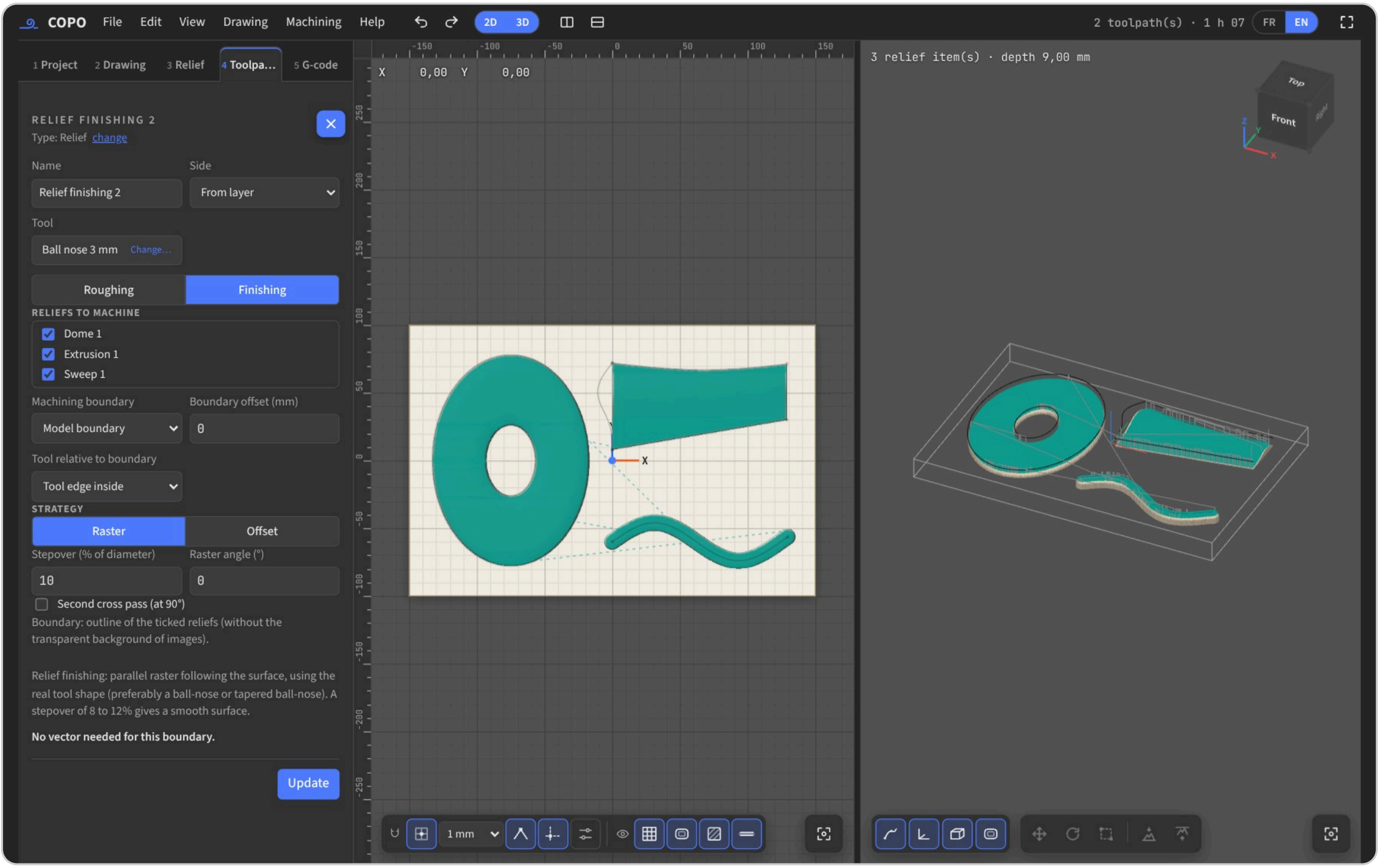
Machining the relief

In the **Toolpaths** tab, the **Relief** type offers two operations, each shown by a small cross-section: roughing as steps around the relief, finishing following the surface. The strategies have their picture too: *Z levels* (flat steps) and *3D raster* for roughing, *Raster* (back and forth) and *Offset* (passes following the outline) for finishing. Below the list of reliefs to machine, the **Advanced options** section (open by default) gathers the machining limit, with a picture showing the chosen limit in blue (model outline, stock outline or selected paths), the offset, the tool position, the strategy and the steps.

- **Roughing**: raster by Z levels with a flat end mill (the largest is suggested), each level stepping down by the pass depth, leaving an *allowance* (0.5 mm by default) for finishing. Default stepover: 40% of the diameter.
- **Finishing**: parallel raster following the surface with the real tool shape, never gouging the relief. Use a ball-nose, a tapered ball-nose (defined by D, R and L) or, for fine details, a flat-tip engraving bit (angle and tip diameter). A round-tip engraving bit is described as a tapered ball-nose, using the box dimensions: shank diameter D, tip radius R, cutting length L; COPO derives the angle (D 6, R 1, L 20: 5.99°). Default stepover: 10% of the diameter; adjustable raster angle and an optional *second cross pass* at 90°.

- **Reliefs to machine**: tick the items whose outline limits machining (all by default). Heights stay those of the full relief.
- **Machining boundary**: *Model boundary* (the ticked reliefs, without the transparent background of images), *Stock boundary* (the whole stock) or *Selected vectors* (closed shapes selected on the workspace). *Tool relative to boundary*: *tool edge inside* (default, the bit does not cross the vector) or *tool center on boundary* (the bit overhangs by one radius, useful to finish a relief edge cleanly). *Boundary offset* enlarges (positive) or shrinks (negative) the boundary.
- **Strategy**: for roughing, *Z levels* (machines only below each level, in runs) or *3D raster* (each line follows the surface without going below the level, fewer retracts); for finishing, *Raster* (parallel lines at the chosen angle, optional cross pass) or *Offset* (contours parallel to the boundary edge, from outside in, climb or conventional).

Passes are chained in zigzags region by region and short links stay on the surface, to limit retracts and air moves. If the relief changes after calculation, the toolpath says so in orange: open it and click **Update**. Finishing files are long (tens of thousands of lines), as in any carving software; the 3D preview replays roughing then finishing.



2 toolpath(s) · 1 h 07

X 0,00 Y 0,00

3 relief item(s) · depth 9,00 mm

RELIEF FINISHING 2

Type: Relief change

Name: Relief finishing 2

Side: From layer

Tool: Ball nose 3 mm Change…

Roughing | Finishing

RELIEFS TO MACHINE

☑ Dome 1
☑ Extrusion 1
☑ Sweep 1

Machining boundary: Model boundary

Boundary offset (mm): 0

Tool relative to boundary: Tool edge inside

STRATEGY

Raster | Offset

Stepover (% of diameter): 10

Raster angle (°): 0

☐ Second cross pass (at 90°)

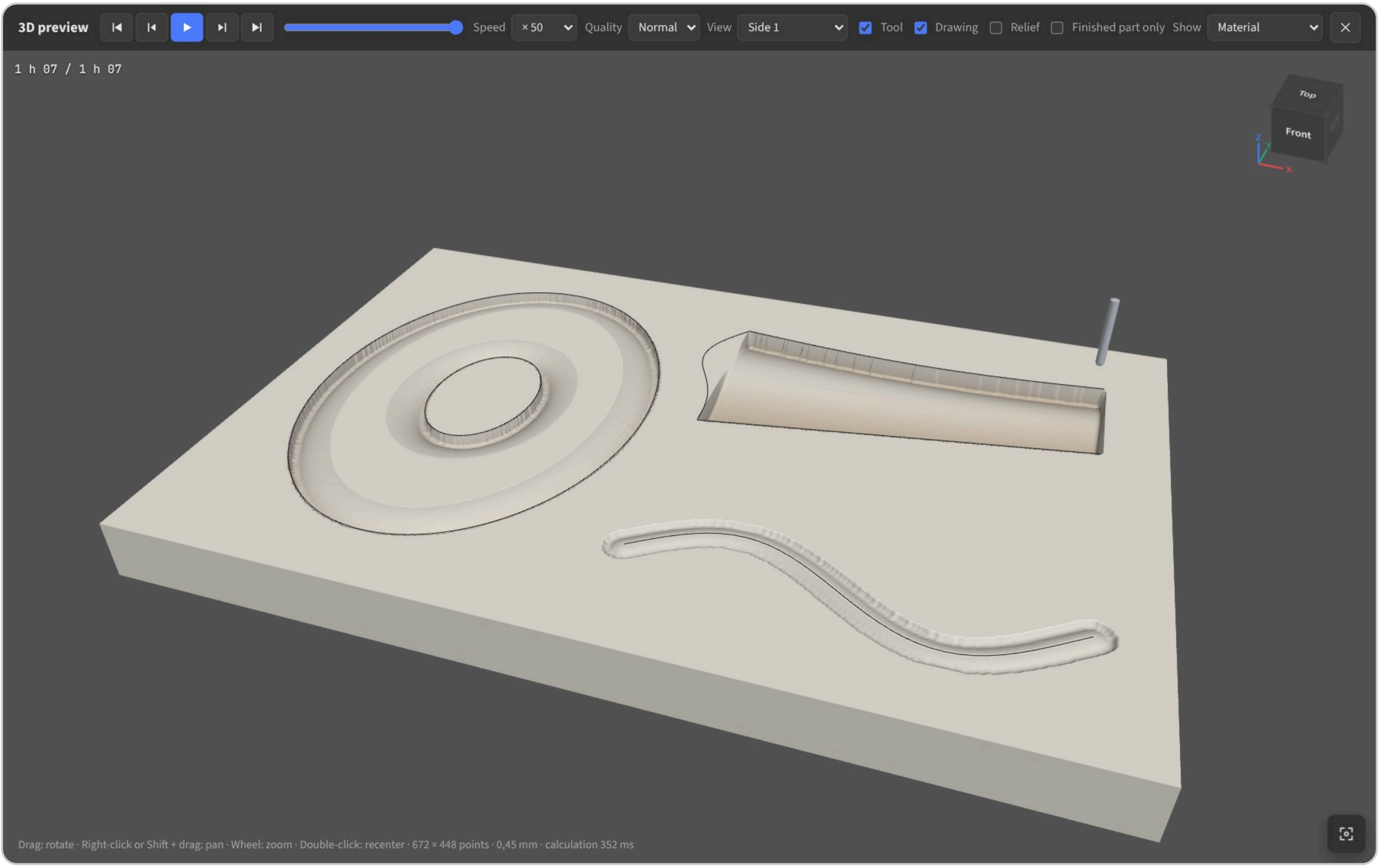
Boundary: outline of the ticked reliefs (without the transparent background of images).

Relief finishing: parallel raster following the surface, using the real tool shape (preferably a ball-nose or tapered ball-nose). A stepover of 8 to 12% gives a smooth surface.

No vector needed for this boundary.

Update

Roughing with a 6 mm flat end mill then finishing with a 3 mm ball-nose, with no boundary selected.



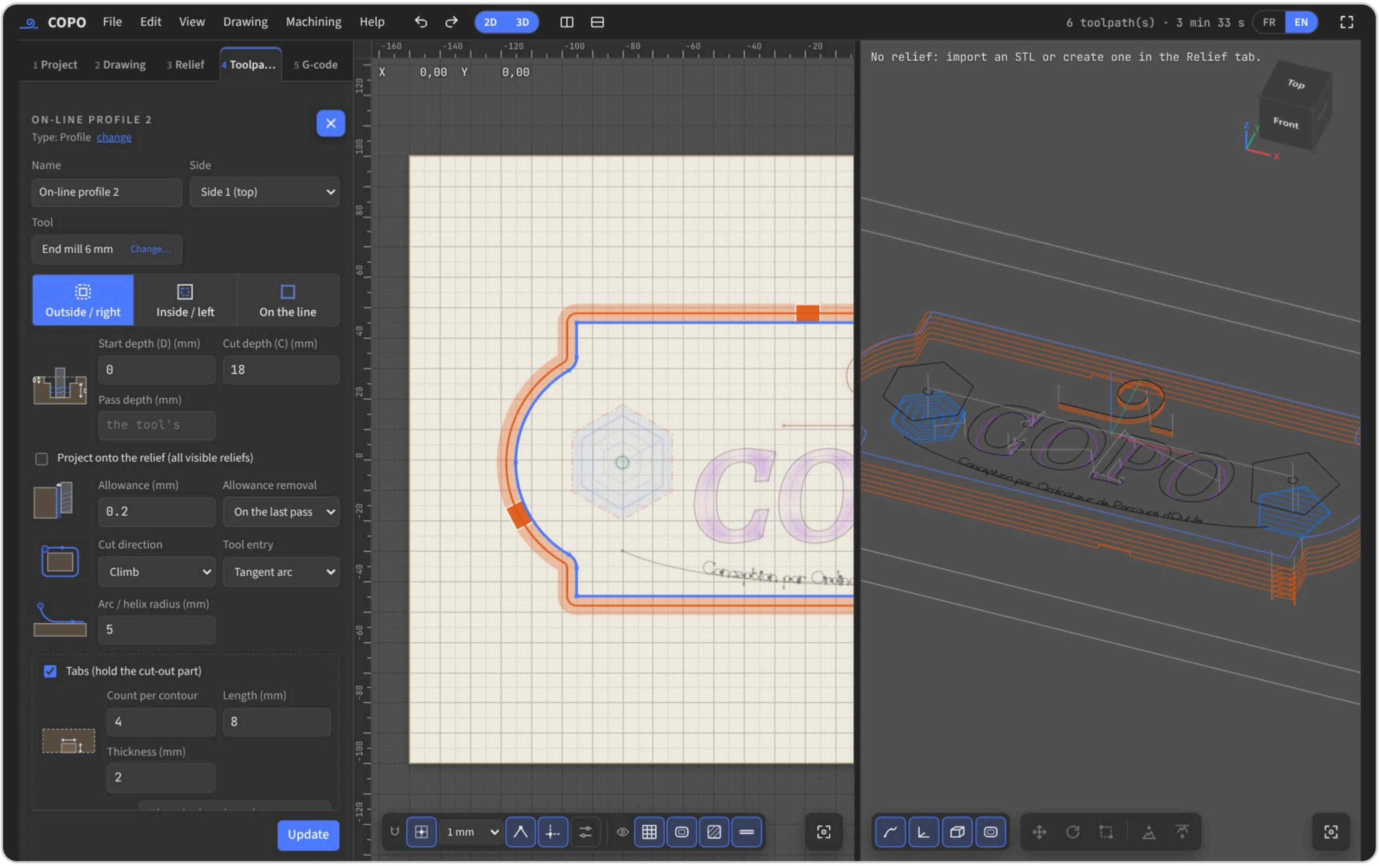
Speed ×50 · Quality Normal · View Side 1 · ☑ Tool · ☑ Drawing · ☐ Relief · ☐ Finished part only · Show Material

1 h 07 / 1 h 07

Drag: rotate · Right-click or Shift + drag: pan · Wheel: zoom · Double-click: recenter · 672 × 448 points · 0,45 mm · calculation 352 ms

The result in the 3D preview.

Toolpaths



6 toolpath(s) · 3 min 33 s

ON-LINE PROFILE 2
Type: Profile change

Name: On-line profile 2
Side: Side 1 (top)
Tool: End mill 6 mm
Start depth (D) (mm): 0
Cut depth (C) (mm): 18
Pass depth (mm): the tool's
☐ Project onto the relief (all visible reliefs)
Allowance (mm): 0.2
Allowance removal: On the last pass
Cut direction: Climb
Tool entry: Tangent arc
Arc / helix radius (mm): 5
☑ Tabs (hold the cut-out part)
Count per contour: 4
Length (mm): 8
Thickness (mm): 2

X 0,00 Y 0,00

No relief: import an STL or create one in the Relief tab.

Click a toolpath in the list to edit it: its path is highlighted on the canvas. The text under the form sums up what the machine will do.

At the top of the tab, the four buttons **Profile**, **Pocket**, **V-carve** and **Drilling** create a toolpath: select the shapes on the canvas (even with the settings open), then click the operation. Its settings replace the buttons and the list in the panel, hiding nothing of the canvas or the 3D view; the line *Type: Profile · change* lets you switch operation. Clicking a toolpath in the list opens its settings to edit it. The **blue cross** at the top right, or a right click on empty space in the drawing, closes them and brings the list back.

While the toolpath settings are open, the rest of the drawing and the other toolpaths are **dimmed**: only the selected vectors and the current toolpath stand out. Everything returns to normal when they close. At the bottom of the settings, a **status line** says what will be machined, for example “3 vectors selected (2 closed, 1 open)”; it turns red when nothing is selected, and warns when a pocket, V-carving or drilling will ignore open vectors.

The four types

- **Profile**: *Outside / right* (the part is kept), *Inside / left* (the hole is cleared) or *On the line*. On an open line, Outside runs the tool to the right of the path and Inside to the left, in its drawing direction (Drawing › Reverse direction to change it); “On the line” follows the stroke or runs left or right according to the “Open lines” setting. On an open line offset to the right or left, ramp, tangent arc and ramped arc entries apply too: the arc enters and leaves on the waste side.
- **Pocket**: four strategies, on one row of icons: *Spiral in → out* (starts in the middle and widens to the edge), *Spiral out → in* (follows the edge first, then tightens), *Zigzag* (parallel lines at the chosen angle, never a pass around the edge: edges slightly scalloped; the allowance does not apply) and *Zigzag + contour* (the same lines, then one pass around the edge to clean it). With a spiral, if a strip too wide for a single pass is left in the middle (depending on the pocket width and stepover), a clean-up pass is added automatically. *Rest machining* uses a smaller bit for the corners a larger bit could not reach: keep the same depth and place this toolpath after the original pocket.
- **V-carving**: the V-bit follows the medial axis of the path; it goes deeper where the stroke is wide and rises up to the tip of every corner. A maximum depth gives a flat bottom. The *pass depth* (the tool’s when left empty) splits the carving into steps, as in Aspire: deep areas are cut in several goes, with the same final result. The *calculation precision* (0.1 mm by default) is the spacing of the points used to compute the medial axis: 0.05 mm for small lettering. A *clearance tool* (flat end mill) can clear the flat areas: COPO then adds a linked “clearance” toolpath just before the V-carve (set it up from the V-carve; it goes away when the tool is removed), and the V-bit only cleans up the edge.
- **Drilling**: one hole at the center of each closed shape, with optional pecking and dwell at the bottom (0 = none). **Pecking** is filled in automatically from the tool, set in the tool library according to the material: aluminium, copper, brass (long chips) 0.5 × D; steels, cast iron and plastics 1 × D; wood and MDF 2 × D. With a flat end mill, circles larger than the bit can be milled helically to the drawn diameter (box unticked by default); the option is hidden for a drill bit, a ball-nose or a V-bit. When it is unticked, an orange warning points out circles larger than the bit, which will be drilled at the bit diameter (handy when you draw large circles just to see and select the holes more easily). With a drill bit, each hole is drilled at the bit diameter; COPO shows the tip length to add for a through hole.
- **Threading**: tapped holes and screws milled with a *thread mill* (single-form, 60°), centered on each selected circle. Set internal (tapped hole) or external (screw), pitch, nominal diameter (0 = the circle’s: draw an 8 mm circle for M8), thread hand and number of radial passes; the thread length is the cut depth.

COPO shows the pre-drill diameter (nominal Ø − pitch, e.g. Ø6.75 for M8 × 1.25). The bit always climb-mills, one helix turn per pitch, with tangent half-circle lead-in and lead-out; the feed entered is the cutting-edge feed and COPO corrects it at the tool center. For a screw, clear the material around the cylinder first. Two thread mills (Ø4 and Ø8 mm) were added to the library with estimated settings.

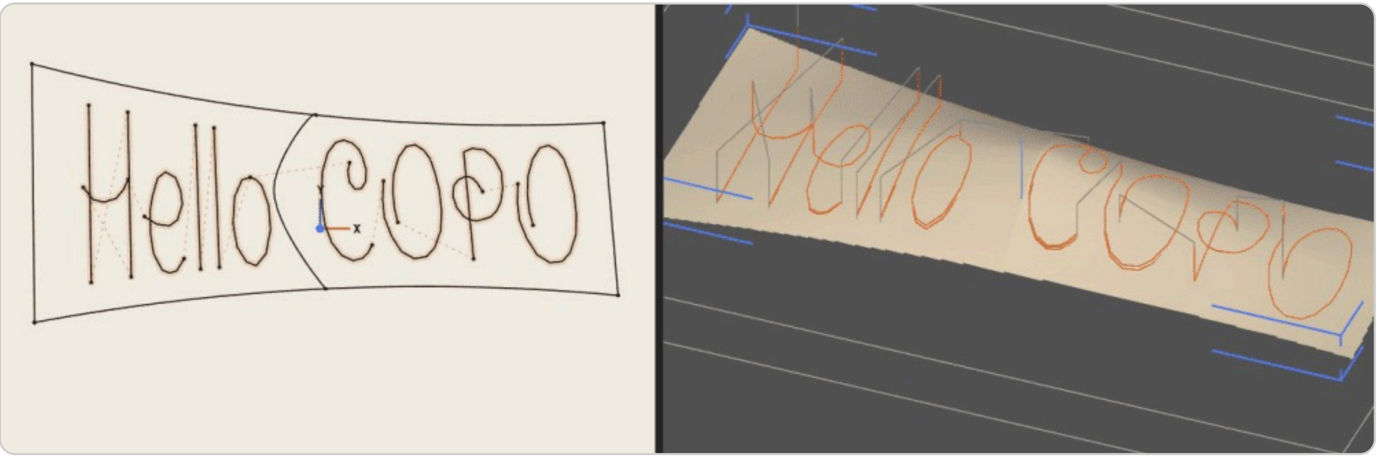
Toolpath recomputation: when the drawing changes, COPO only recomputes the toolpaths concerned (those whose shapes or settings changed); undo and redo reuse calculations already made. A long calculation shows a progress bar. In **Settings** › **General**, recomputation can be set to *Manual*: affected toolpaths are marked ⟳ and keep their old path until you click **Recompute** (below the toolpath list); G-code export reminds you. The **Recompute all toolpaths** button stays available in automatic mode.

Advanced options: below the cut direction, this folding section (closed by default, COPO remembers your choice) gathers the allowance and its removal, projection onto the relief, the tool entry and ramp length, rest machining (pocket), and for drilling the helix, pecking and dwell. The allowance is 0 by default.

Passes, allowance and cut direction

Start depth (D) and **Cut depth (C)**, as in Aspire: machining starts at depth D below the surface (for example at the bottom of a pocket already cut) and goes down by C from there, so the bottom is at D + C. Leave D at 0 to start from the surface. In V-carving, the pattern is computed as if the surface were at D, which is what inlays use. The small diagram next to the fields shows what D and C are. A *negative* value starts above the surface: the first passes cut air, useful when the stock is thicker than planned or its surface uneven (D = −2 and C = 5: start 2 mm above, bottom 3 mm deep).

Project onto the relief (profile, pocket, V-carving): the toolpath follows the surface of all visible reliefs of the Relief tab instead of the flat stock top (no need to select the relief: only select the toolpath's vectors); depths D and C are then measured from the relief. The contact height takes the cutter shape into account (V, ball-nose, flat), to carve a pattern or lettering on a domed surface. Outside the relief, the toolpath stays measured from the stock top. Side 1 only; if the relief changes, the list flags that the toolpath must be recalculated.



Single-line text engraved *on* a sweep: the toolpath (orange) follows the relief surface.

The cut depth is split into passes of the chosen depth. An **allowance** leaves a little material on the walls while roughing; it is removed on the last pass or by a full-depth finishing pass. The cut direction is *climb* or *conventional*.

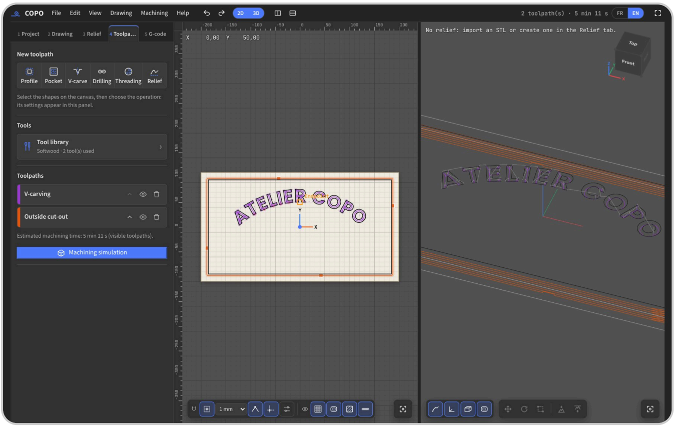
Tool entry

ENTRY	BEHAVIOUR
Back-and-forth ramp	The tool goes down while moving, returns to the start at full depth, then goes around. Each pass starts and ends at the same point. Default setting.
Continuous ramp	The tool goes down while moving and continues around; the ramp portion is recut at the end.
Tangent arc, ramped arc	Arc lead-in and lead-out on the contour, to avoid a mark on the finished wall.
Helix	The tool spirals down, ideal for pockets and holes.
Straight plunge	Vertical plunge, for center-cutting bits or V-carving.

Tabs

Tabs hold a cut-out part until the end. Set their count per contour, length and thickness, flat or triangular (easier to break). **Place by hand on the canvas…** lets you add a tab with a click on the toolpath, drag it or remove it.

Text and V-carving


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
1 Project
2 Drawing
3 Relief
4 Toolpa…
5 G-code
New toolpath
Profile
Pocket
V-carve
Drilling
Threading
Relief
Tools
Tool library
Softwood · 2 tool(s) used
Toolpaths
V-carving
Outside cut-out
Machining simulation
ATELIER COPO

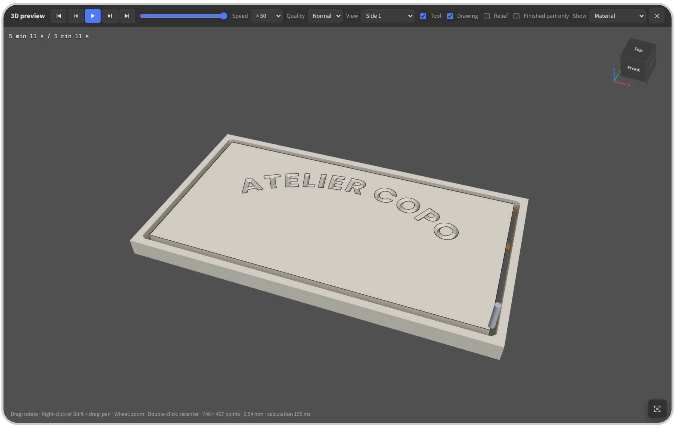
X 0,00 Y 50,00

Select the shapes on the canvas, then choose the operation: its settings appear in this panel.

Estimated machining time: 5 min 11 s (visible toolpaths).

No relief: import an STL or create one in the Relief tab.

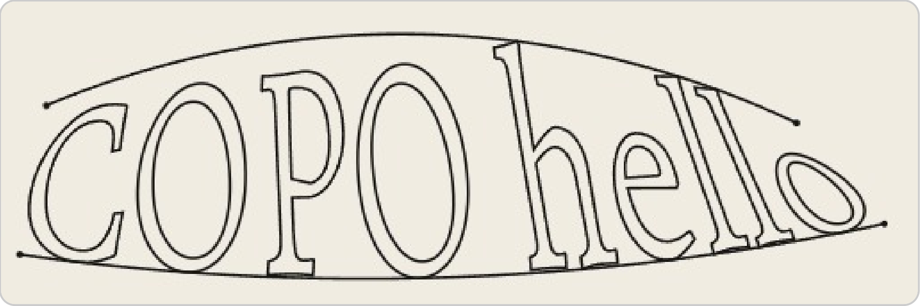
Text placed above an arc: every setting of the panel (placement, alignment, letters, spacing…) changes it live.


3D preview
Speed
Quality
Normal
View
Side 1
Tool
Drawing
Relief
Finished part only
Show
Material
ATELIER COPO

5 min 11 s / 5 min 11 s

The same job in the 3D preview.

Choose a font in the **Text** block list: installed fonts (Chrome and Edge) or your fonts folder (**Open a fonts folder…** button of the block; the ⓘ icon explains where to keep them), with a preview. Two boxes at the top of the list, *COPO fonts* and *Computer fonts*, shorten the list; the choice is remembered. Take the **Text** tool T: a dashed outline follows the cursor and shows where the text will go. The **placement** is chosen on five pictures: *Straight* (click on the canvas), *Above the curve*, *On the curve* or *Below the curve* (click a line, an arc, a circle, an outline…) and *Between two curves*; a blue box says what to click. On a curve, set the **alignment** (left, centre or right of the curve; for a closed outline, relative to the clicked point), the **letters** (aligned to the curve or kept upright), *Text on the other side of the curve* (to write inside an arc, for example) and the **spacing** between letters (also for straight text). **While the Text tool stays active, every setting changes the text you have just placed, live**: text, font, height, placement, alignment… no more deleting and starting again. To edit a text later, select it: the **T** icon of the Selection block reopens its settings. The text becomes grouped vectors, ready for V-carving, pocketing or profiling. In **Between two curves** placement, click the bottom curve (the baseline) then the top one: the letters stretch to fill the space between them, like an envelope (two arcs for an arched sign, for example). The *offset* moves the text away from the curve (on a curve) or keeps a margin with each of the two curves (between two curves).



Between two curves text: the letters stretch to follow the bottom and top curves.

Single-line fonts

COPO comes with about sixty **single-line fonts** (Hershey, EMS, Cutlings, Relief SingleLine, Routed Gothic…), stored in the `fonts/trait-unique` folder and listed first in the font list, with a preview. Each letter is a single stroke: the text is engraved in one pass with a **Profile On the line** toolpath, with a V-bit or a small end mill. It is the fastest way to mark part numbers, plates and assembly marks. When the selection holds only open strokes, **Machine…** offers this profile directly. The `fonts/apercu-polices.html` catalogue (link “See the single-line font catalogue” in the Text block) shows them all with a preview of your text. You can also add your own SVG fonts with **Add fonts…**.

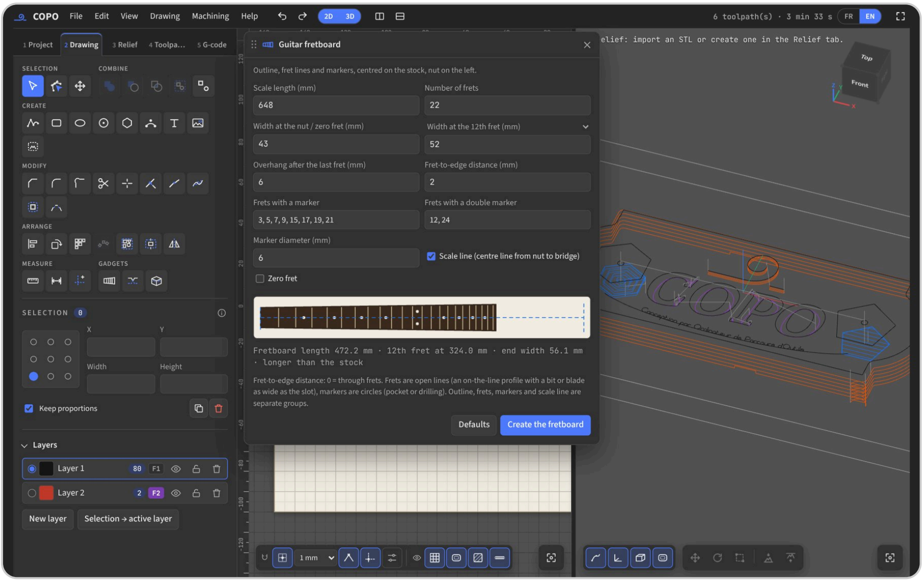
Decorative fonts

Nine **decorative font** families are also bundled, in the `fonts/decoratives` folder, and listed under *Decorative fonts (bundled with COPO)*: **URW Chancery** (chancery italic), **Alegreya** (roman and italic), **Cinzel Decorative** (ornate capitals, 3 weights), **Pirata One** (blackletter), **Chicle**, **Comic Neue** (6 styles), **Protest Riot**, **Great Vibes** (script) and **Smokum** (western). These are outline fonts: machine them with V-carving, a pocket or a profile. For thin scripts such as Great Vibes, use a large enough text height (20 mm or more) so the bit fits in the hairlines. The *COPO fonts* box hides them together with the single-line fonts. Licences: SIL Open Font License for most, Apache 2.0 for Smokum, AGPL 3 with font exception for URW Chancery (see `fonts/decoratives/LICENCES.txt`).

Lutherie: fretboard and inlay

Guitar fretboard

Extensions › Guitar fretboard… draws the fretboard, nut on the left, centred on the stock. Set the **scale length**, the **number of frets**, the **width at the nut** and the second width, *at the 12th fret* or *at the end of the scale*, the overhang after the last fret, the **frets with a marker** and **a double marker** (lists of numbers) and the marker diameter. The **scale line** (ticked by default) draws the centre line from the nut to the bridge, with a mark at each end, to place the bridge. With the **Zero fret** box, a fret 0 is drawn at the start of the scale (with the same edge distance as the others) and the board starts before it, at the distance set in *Board start → zero fret*; the “nut” width is then the zero-fret width. The **fret-to-edge distance** stops the fret lines short of the board edge (0 = through frets). The preview updates live. The window opens next to the left panel, like the toolpath settings, and moves by its title.


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
6 toolpath(s) · 3 min 33 s
FR
EN
1 Project
2 Drawing
3 Relief
4 Toolpa…
5 G-code
SELECTION
COMBINE
CREATE
MODIFY
ARRANGE
MEASURE
GADGETS
SELECTION 0
X
Y
Width
Height
Keep proportions
Layers
Layer 1
Layer 2
New layer
Selection → active layer
1 mm
Top
Front

elief: import an STL or create one in the Relief tab.

Guitar fretboard

Outline, fret lines and markers, centred on the stock, nut on the left.

Scale length (mm): 648
Number of frets: 22
Width at the nut / zero fret (mm): 43
Width at the 12th fret (mm): 52
Overhang after the last fret (mm): 6
Fret-to-edge distance (mm): 2
Frets with a marker: 3, 5, 7, 9, 15, 17, 19, 21
Frets with a double marker: 12, 24
Marker diameter (mm): 6
☑ Scale line (centre line from nut to bridge)
☐ Zero fret

```
Fretboard length 472.2 mm · 12th fret at 324.0 mm · end width 56.1 mm
· longer than the stock
```

Fret-to-edge distance: 0 = through frets. Frets are open lines (an on-the-line profile with a bit or blade as wide as the slot), markers are circles (pocket or drilling). Outline, frets, markers and scale line are separate groups.

Defaults
Create the fretboard

The **Guitar fretboard** window with its preview: board, frets, markers and the dashed scale line.

Fret positions

COPO uses the exact equal-temperament formula, the one used by reference fret calculators:

$$\mathbf{d_n = L - L / 2^{n/12}} \quad \text{i.e.} \quad d_n = L \times (1 - 2^{-n/12})$$

where L is the scale length, n the fret number and d_n the distance from the nut to the fret. Each fret shortens the string by a semitone, a factor of $2^{1/12} \approx 1.05946$; the 12th fret falls exactly at half the scale. The “rule of 17.817” gives the same result; the old, less accurate rule of 18 is not used. No compensation is applied to the frets: intonation is set at the bridge, moved back a few millimetres from the scale line.

FRET	1	5	7	12	
Distance from the nut (648 mm scale)	36.37 mm	163.51 mm	215.64 mm	324.00 mm	466.1

The result is separate groups: the scale line (not to be machined), the outline, the fret lines (open lines, to cut with an **On the line** profile and a bit or blade as wide as the slot) and the markers (circles, pocketed or drilled).

V-carve inlay

Select the design (closed shapes), then **Drawing › V-carve inlay…**. The settings open in the left panel, with a dimensioned diagram as in Aspire: **a** pocket depth, **g** glue gap at the bottom, **b** male height below the backing (it must exceed a − g: the rest is planed off), **c** backing overhang, and a **name** for the toolpaths. Choose the V-bit (**Library…** button for the whole library) and, if you like, a flat **clearance tool** that first clears the flat areas (faster on large designs). The overhang *c* is automatic when left empty; the *Create the toolpaths* box can be unticked to draw only the male piece. COPO creates:

- a *female* V-carving of the design in your part;
- the *male* plug: the mirrored design inside a frame, on the “Inlay · male” layer, with a V-carving that starts at the pocket depth minus the gap and goes down by the surface gap. Its toolpaths are created *hidden*: to cut it from the inlay wood, show them and hide the female ones before saving the G-code.
- with a clearance tool, a “clearance” pocket before each V-carving, where the V would reach its full depth.

Flip the plug, glue it into the female pocket, then plane or sand off the excess. The V-carving **Start depth** setting is also available by hand in the toolpath form.

Pocket inlay (flat end mill)

Select the design (closed shapes), then **Extensions › Pocket inlay…** or, with a right click, **Inlay › Pocket**. The settings open in the left panel. Choose the **flat end mill** (the **Library…** button gives access to the whole tool library), the **glue gap** (0.1 mm by default) and how to make the male piece (*Male piece on a backing board* box). A small dimensioned diagram, as in Aspire, shows the values to fill in: **a** pocket depth, **b** male height (below the backing, or piece thickness without backing) and **c** backing overhang (automatic when left empty). The *Create the toolpaths* box (ticked by default) can be unticked to create only the outlines. For large pockets, tick *Rough out with a bigger cutter*: COPO recommends roughing or not depending on the pocket width and the main cutter, and suggests a cutter (from the project, otherwise from the library). The big cutter then clears the pocket and the plug background, and the main cutter only cleans the corners and narrow passages (rest machining): on a 60 × 40 mm pocket, the time drops by almost three.

A flat end mill cannot cut a corner sharper than its radius. COPO therefore computes a **common outline** whose corners, inside and outside, are all rounded to at least the cutter radius: the pocket and the insert are machined exactly on this outline and fit without touch-up. The preview shows this outline in blue, the original drawing in dashed red, and the share of area changed; a warning flags parts thinner than the cutter, which disappear.

- **Male piece on a backing board** (box ticked, default): the design is machined mirrored, raised, in a thick board, to a height *b* greater than the pocket *a*: once glued, the board stays b − a above the part, leaving room for the saw cut. Glue the board upside down into the pocket, then saw or plane off the excess. All parts of the design stay joined and go in at once.
- **Male piece without backing** (box unticked): the design is cut out of a thin board of thickness *b*, with holding tabs. The *cut-out depth* is b when left empty; use a little more (b + 0.2 mm, say) to cut cleanly through if your spoilboard allows it. Make b equal to a for a flush piece, a little more if you want to sand after gluing; COPO warns when b is smaller than a. Break the tabs, sand, then glue the piece into the pocket.

Deleting an inlay toolpath also deletes the outlines it created (your original drawing is kept). COPO creates the pocket on the “Pocket inlay · female” layer and the insert to the right of the design, on the “Pocket inlay · male” layer. Both toolpaths are visible and go into the G-code; hide one (eye) if you machine the pieces separately.

Finger-joint box

The **Extensions** › **Finger-joint box…** menu opens this module, which draws a box assembled with straight finger joints, like Vectric's “Box Creator”. Set the **outer dimensions** (width X, depth Y, height Z), the **material thickness** (the stock's by default), the **finger width** (their number, always odd, follows), the **joint allowance** (0.1 to 0.2 mm in wood), the **cutting bit**, the **top joint** (*finger-jointed* for a closed box, the default, or *laid flat* for a removable lid) and the **faces to machine** (Top, Bottom, Sides, Ends). Fingers are spread evenly between the corners: the corner zone, as thick as the material, only exists where another panel joins, so an open box keeps the same pitch up to the top. Fingers may be narrower than the thickness; COPO warns when the slots become narrower than the bit. The *Dogbone reliefs* option adds an arc the size of the bit at the bottom of each slot so square fingers fit.

The preview shows the panels laid flat. **Create the box** places them on the stock, spaced for the bit: cut them with an outside profile and tabs. The panel volumes fit exactly: no overlap or gap at the corners.

Rotary graduations

Rotary graduations window: 36 divisions, medium lines every 5 and long lines every 10, preview on the unrolled sheet and lines wrapped in 3D.

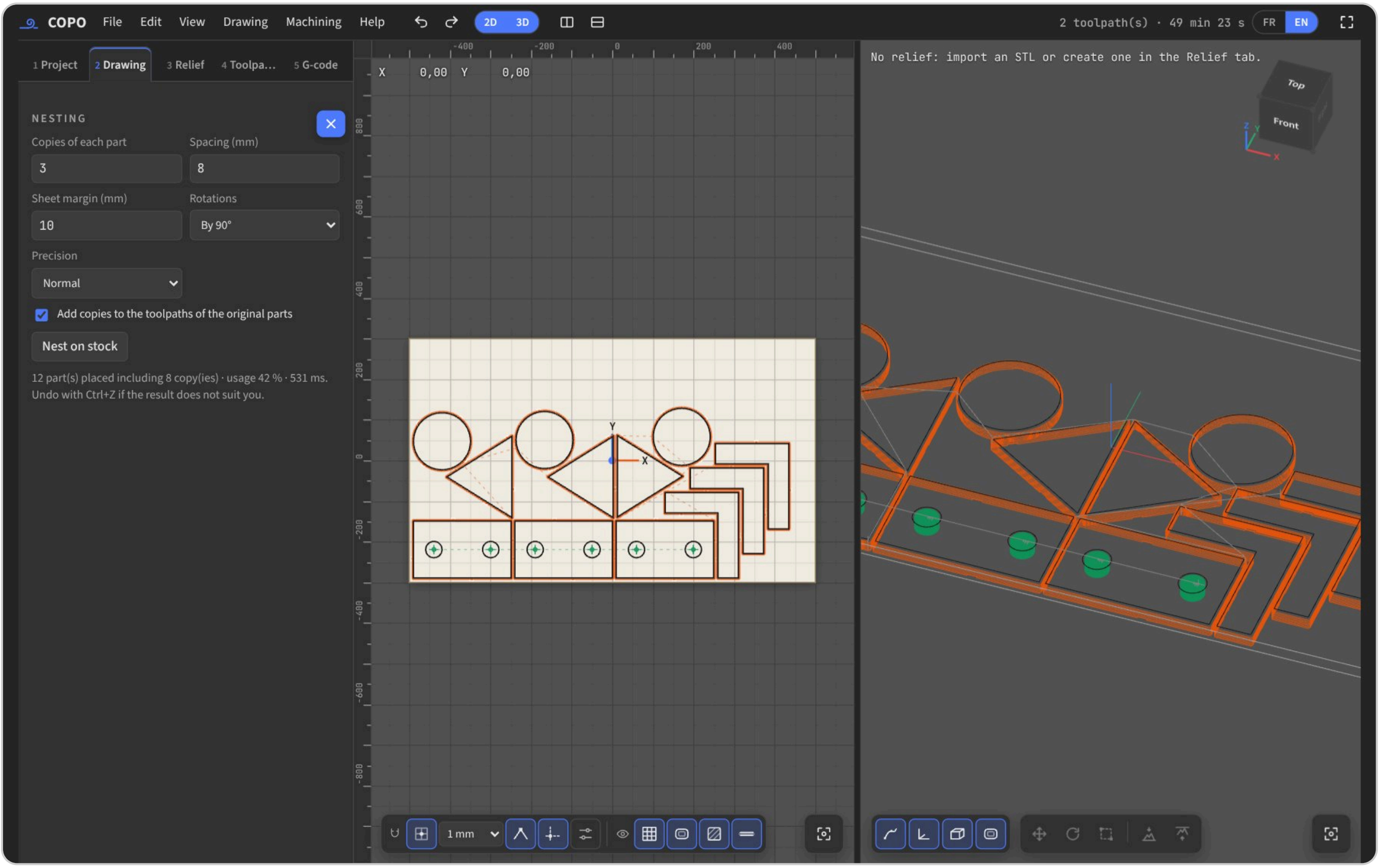
36 divisions around a Ø 90 mm cylinder: a medium line every 5, a long one every 10.

In a **rotary** project, **Extensions › Rotary graduations…** draws evenly spaced lines around the cylinder: graduations, verniers, flutes, gear teeth. The lines are always **parallel to the cylinder axis** chosen in the Project tab (horizontal when the cylinder lies along X, vertical along Y) and spread around the circumference (π × D).

- **Number of divisions** and **span**: at 360°, as many lines as divisions all the way round. Below 360°, the divisions cover the span with a line at each end; for a tenth vernier, for example, 10 divisions over 9°.
- **First line angle**: 0° is the origin line, like the 0°, 90°, 270° marks on the plan.
- **Position and length**: as in the Selection block, choose the line's reference point (start, middle or end), then its X (or Y) position relative to the origin and its length.
- **Longer lines**: a medium line every *n* lines and a long line every *m* lines, each with its own length; they start from the same reference point. *First line number* shifts this count.

The preview shows the unrolled sheet and the pitch, in degrees and in millimetres on the surface. **Create the lines** places them as a group on the active layer; machine them with V-carving or an “on the line” profile.

Nesting



2 toolpath(s) · 49 min 23 s

X 0,00 Y 0,00

No relief: import an STL or create one in the Relief tab.

NESTING

Copies of each part: 3

Spacing (mm): 8

Sheet margin (mm): 10

Rotations: By 90°

Precision: Normal

☑ Add copies to the toolpaths of the original parts

Nest on stock

12 part(s) placed including 8 copy(ies) · usage 42 % · 531 ms.
Undo with Ctrl+Z if the result does not suit you.

Four parts in three copies on a 1000 × 600 mm sheet. The L shapes interlock and the copies join the toolpaths of the original parts.

Drawing › Nesting… opens the block in the Drawing tab. Select the parts (with no selection, the whole drawing is used) and set the number of copies, the spacing, the sheet margin, the allowed rotations and the precision. COPO places the parts following their true shape, largest first, as far down and left as possible. A group or a text stays in one block. Parts that do not fit stay selected, and the sheet usage is shown.

> For cut-outs, use a spacing at least equal to the bit diameter. `Ctrl` + `Z` undoes the nesting if you don't like the result.

Double-sided machining

1. Put the vectors of the bottom side on a layer set to **F2** (F1/F2 button in the layer list). Draw them as seen from above, see-through.

2. Choose the flip: around the X axis (top ↔ bottom) or the Y axis (left ↔ right).
3. Side 2 toolpaths are calculated on the flipped geometry; the cut direction is kept.
4. In the G-code, side 1 is followed by an `M0` pause: turn the part over, set X0 Y0 Z0 again at the same point and resume.

The 3D preview shows both sides; the **Side 2 up** option shows the part flipped.

2D and 3D views

The **2D** and **3D** buttons at the top work as two switches: 2D only (the drawing fills the area), 3D only (the relief), or both at once. When both are shown, the buttons next to them place the views side by side or one above the other, and the **divider** between them can be dragged to enlarge either one (double-click: half and half). The Relief tab shows the 3D view automatically.

The 3D view shows the relief in the stock (wireframe), with the drawing and the axes. The view uses an orthographic projection (no perspective), handy from the front or the side. Hold the right button: orbit (around the point under the cursor, unlimited, you can go under the part); hold the middle button: pan; wheel: zoom towards the cursor; double-click: whole stock; a left click selects. As in Fusion, FreeCAD or Tinkercad; the mouse is set in Settings › 3D views. Top right: the view cube. At the bottom, icon bars like in the 2D view: on the left, show or hide the drawing, the axes, the stock and the toolpaths (cuts in their type's colour, rapid moves in grey); in the middle, the transform tools; on the right, recentre on the relief. The stock and the toolpaths are shown every time COPO opens. If you hide the toolpaths, the one you hover in the Toolpaths tab list, or the one you are editing, stays visible in 3D.

Drawing vectors can also be selected in the 3D view: click a line (the cursor turns into a hand). The selection is shared by both views, in blue; Shift + click adds or removes a vector, a click on empty space deselects. Handy for picking the outline of an STL model before creating its toolpath.

When COPO opens, the 2D canvas and the 3D view are shown side by side. The 3D view also opens by itself with the Relief and Toolpaths tabs. If you closed it with the **3D** button, COPO respects that choice until the page is reloaded.

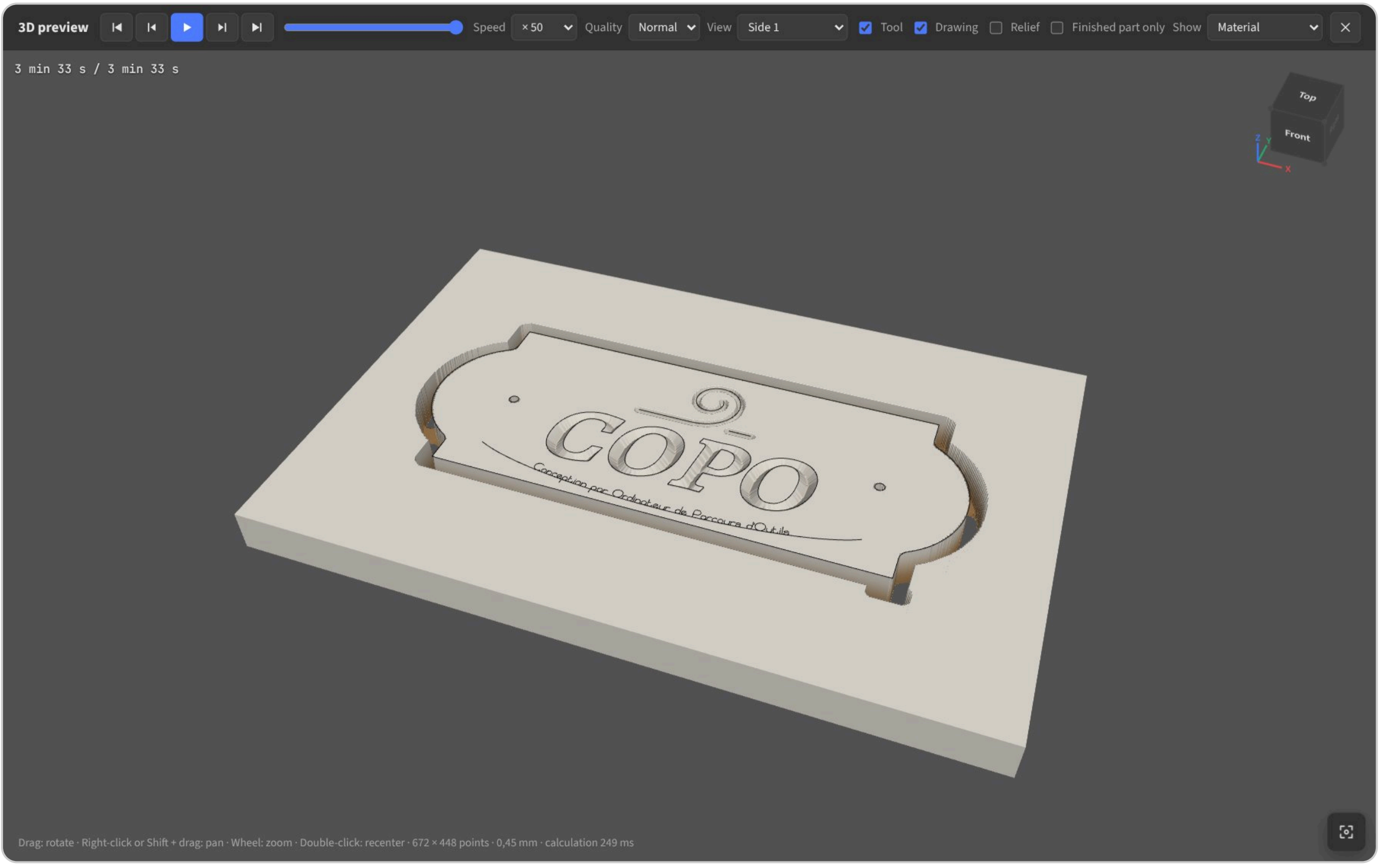
Transforming a relief in 3D. Click an item in the 3D view to select it (its box corners are highlighted; the selection is shared with the 2D view and the list). The bottom bar offers three tools, with dimensions shown at the top left:

- **Move**: drag the X, Y or Z arrows (Ctrl: whole millimetres). In Z, the relief top snaps to the stock top and its bottom to the stock bottom; in X and Y, its edges

snap to the stock edges.
- **Rotate**: drag the blue ring (around Z); for an STL, the red and green rings also tilt it around X and Y (Ctrl: 15° steps).
- **Resize**: drag a corner to scale keeping proportions (Shift, or “Keep proportions” unticked: freely), a side to change only the width or the length, or the blue dot above for the height: the top moves, the bottom stays put. Ctrl moves in whole millimetres.
- The last two buttons **drop the relief onto the stock bottom** or **align its top with the stock top**.

Each transformation can be undone with **Ctrl+Z**; the relief panel keeps the exact values.

3D preview



Speed ×50 · Quality Normal · View Side 1 · ☑ Tool · ☑ Drawing · ☐ Relief · ☐ Finished part only · Show Material

3 min 33 s / 3 min 33 s

Drag: rotate · Right-click or Shift + drag: pan · Wheel: zoom · Double-click: recenter · 672 × 448 points · 0,45 mm · calculation 249 ms

Hold the right button to orbit, the middle button to pan, wheel to zoom, double-click to recenter.

Detail: the simulation only computes the area actually machined (the rest of the stock is drawn solid); the smaller the area, the finer the result. The *Resolution* setting (Standard, High or Very high: 0.5, 1.5 or 3.5 million points) raises the detail further; the resolution used is shown at the bottom of the window.

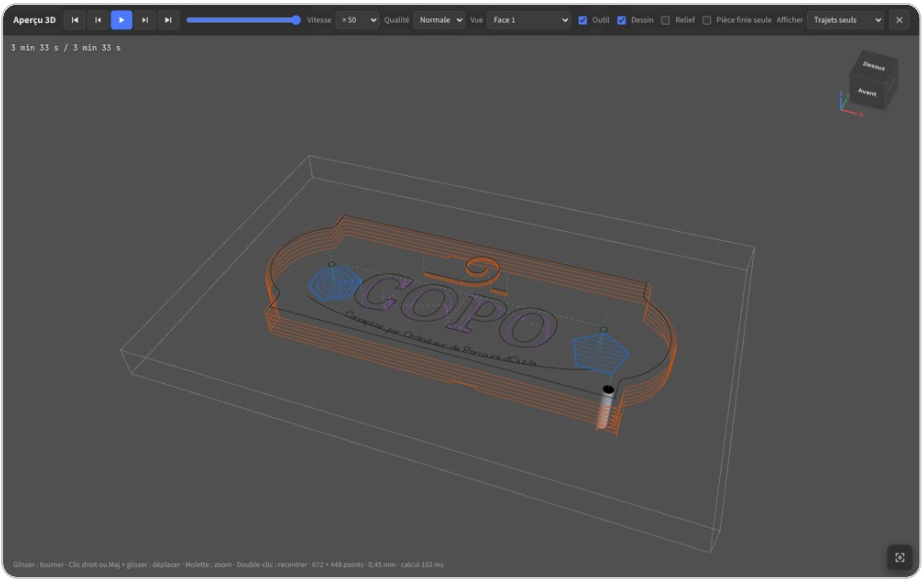
The preview simulates material removal through the real stock thickness, through-holes included. The playback buttons replay the machining: **back to start**, **step back**, **play / pause**, **step forward** and **go to end**; one step is one tool move. Keyboard: left and right arrows, Home, End and Space. The slider jumps to any moment. Elapsed and total time show at the top left of the view; the recentre button is at the bottom right, as in the other views, and the cross at the top right closes the preview.

View cube (top-right corner): it turns with the view and shows the faces (Top, Front, Right…) and the X, Y, Z axes. Click a face for that view (Top = plan view, like the drawing), or a corner for an isometric view. The tool shown keeps its shape: flat, ball-nose, V-bit, drill or thread mill (60° tooth at the tip, relieved neck). Threads appear in the preview as the last pass progresses (ISO 60° profile), inside the hole or around the screw.

Relief and finished part: the **Relief** box overlays the designed relief (a line mesh at its real height) to compare it with the machined material. The **Finished part only** box hides the offcuts: material linked to the stock edge without crossing a full-thickness cut disappears, leaving only the cut-out part(s), as they come off the machine. Tabs do not link the part to the offcut (a small stub stays visible). If no part is cut out, the whole stock stays shown.

Toolpaths in 3D

The **Show** list in the preview bar offers *Material*, *Material + paths* or *Paths only*. The **Drawing** box shows or hides the drawing's vectors, in their layer colour, on the face turned upwards: face 1 vectors in the Face 1 view, face 2 vectors in the Face 2 view. Toolpaths are drawn as lines in space, in the colour of their type; rapid moves (G0) are dashed grey. During playback they are drawn as the tool moves. With a toolpath selected in the list, the **Selected toolpath only** box shows just that one. Great for checking passes, ramps, helixes and clearance heights before cutting.



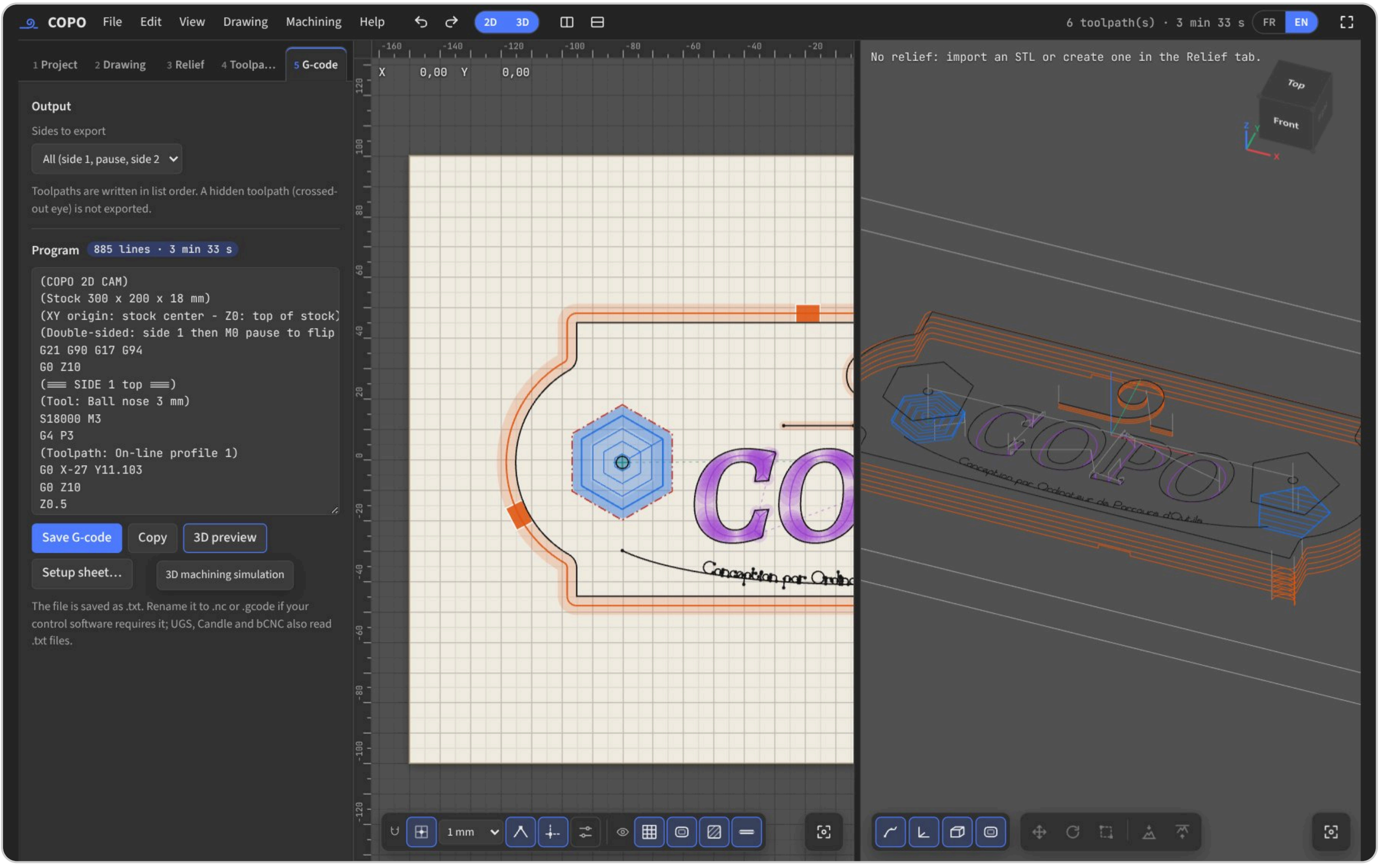
Vitesse ×50 · Qualité Normale · Vue Face 1 · ☑ Outil · ☑ Dessin · ☐ Relief · ☐ Pièce finie seule · Afficher Trajets seuls

3 min 33 s / 3 min 33 s

Glisser : tourner · Clic droit ou Maj + glisser : déplacer · Molette : zoom · Double-clic : recentrer · 672 × 448 points · 0,45 mm · calcul 152 ms

Paths only mode: every cut-out pass is visible, rapids are dashed.

G-code


COPO
File
Edit
View
Drawing
Machining
Help
2D
3D
6 toolpath(s) · 3 min 33 s
FR
EN
1 Project
2 Drawing
3 Relief
4 Toolpa…
5 G-code
X 0,00 Y 0,00
1 mm
Top
Front

Output

Sides to export: All (side 1, pause, side 2

Toolpaths are written in list order. A hidden toolpath (crossed-out eye) is not exported.

Program 885 lines · 3 min 33 s

```
(COPO 2D CAM)
(Stock 300 x 200 x 18 mm)
(XY origin: stock center - Z0: top of stock)
(Double-sided: side 1 then M0 pause to flip
G21 G90 G17 G94
G0 Z10
(══ SIDE 1 top ══)
(Tool: Ball nose 3 mm)
S18000 M3
G4 P3
(Toolpath: On-line profile 1)
G0 X-27 Y11.103
G0 Z10
Z0.5
```

Save G-code
Copy
3D preview
Setup sheet…
3D machining simulation

The file is saved as .txt. Rename it to .nc or .gcode if your control software requires it; UGS, Candle and bCNC also read .txt files.

No relief: import an STL or create one in the Relief tab.

Toolpaths are written in list order; a hidden toolpath is not exported.

POST-PROCESSOR	TOOL CHANGE
GRBL	`M0` pause between two tools.
GRBL manual	Before each tool, the first one included: retract with `G53 G0 Z-10` (adjustable) and `M0` pause. Also a `G53` retract at the end of the program.
Carbide Motion, Onefinity	`M6 T…`: the control software asks you to change the tool.
LinuxCNC, Mach3/4, UCCNC, MASSO	`T… M6` with the tool number, `G40 G49 G80` header, arcs with `G91.1`. `.ngc` file (LinuxCNC), `.tap` (Mach, UCCNC) or `.nc`.
Custom	You write the header, tool change and program end, with variables: `{UNITS}`, `{SAFEZ}`, `{TCZ}`, `{T}`, `{S}`, `{D}`, `{TOOL}`. File extension and `G91.1` are up to you.

Other machines (.pp format). The list also offers, in a separate group, industrial and special machines: Advanced Laser, Axyz, Bridgeport EZTRACK, Busellato GEN J6000, DeskCNC, Haas (with or without arcs), ISEL NCP, Kuka (robot), Lasercomb, Mach2/3 ATC, Biesse Rover, Pacer, Precix, ShopBot (routing, head 2, GC and PRS air drills), Torchmate 2, Turbo CNC, Vision Series 3, WaterJet PRO, rotary-axis machines (Mach2/3 Wrap, CAMaster, CNC Shark, WinCNC, WinPC-NC, ShopBot Indexer), an HPGL plotter / vinyl cutter and a readable text listing of the toolpath (for checking). They follow the logic of Aspire and VCarve .pp post-processors: each sets its units (mm or inches), number format, line numbering, header, tool change, program end and file extension. If your machine is not there and you have its .pp file (PostP folder of Aspire, VCarve or Cut2D), the **Import a post-processor (.pp)…** button loads it: it joins the “Imported” group, is kept in your browser and saved with the project. The usual blocks are handled (header, end, tool change, new toolpath, rapids, feeds, plunges, retracts, IJ or R arcs, helical arcs, dwell, laser / plasma / water jet on and off, feed change), as well as the usual variables: `[TOOL_DIA]` cutter diameter, `[TOOLNAME]`, `[DATE]`, `[TIME]`, stock size (`[XLENGTH]`, `[XMIN]` …), liste des parcours et des outils (`[TOOLPATHS_OUTPUT]`, `[TOOLS_USED]`). COPO lists the blocks it ignores. Post-processors that turn the tool on at the plunge (laser, water jet) need toolpaths with a “Straight plunge” entry: COPO reminds you in the Machine block and the G-code tab. Always do a dry run with a post-processor you have not used yet.

```
(Load tool: End mill 6 mm D6 - set Z zero again then resume)
G21 G90 G17 G94
G53 G0 Z-10
M0
S18000 M3
G4 P3
G0 X10 Y10
G0 Z10
…
M5
G53 G0 Z-10
G0 X0 Y0
M30
```

The post-processor, rapid speed (for the time estimate), safe Z, clearance Z, comments and G2/G3 arcs are set in the Project tab, Machine block. The file gets the extension the post-processor expects (`.nc`, `.tap`, `.ngc`).

Curves and G2/G3 arcs. COPO does not send Bézier curves to the machine as they are: toolpaths are computed as short segments, then, when “G2/G3 arcs” is ticked, circular portions are recognised and written as true G2/G3 arcs (shorter file, smoother motion). A circle drawn with the Circle tool or a vector smoothed in *Arcs* mode gives exact arcs; a free-form curve is approximated by a series of arcs and lines, within tolerance.

> Always check the G-code with a dry run, tool above the stock, before cutting.

Units: millimetres or inches

In the Project tab, Units block, choose the **Length units** (millimetres or inches) and the **Feed units** (mm/min, mm/s or in/min). The choice applies everywhere: drawing, rulers, grid (1/16, 1/8, 1/4 and 1 inch), coordinates, dimensions, tools, toolpaths and setup sheet. In inches, one option shows values as fractions (to 1/64) and another writes the G-code in inches (`G20`) for controllers set up that way.

The project is always saved in millimetres: you can switch units at any time without distorting anything, and a project opens correctly for someone working in the other unit. The choice is remembered in the browser.

Setup sheet

The **Setup sheet…** button in the G-code tab (also in the Machining menu) prepares a two-page setup sheet for the workshop. Enter the part name, the material and your notes; the sheet contains:

- the stock, X0 Y0 origin, Z zero, flip and post-processor;
- a summary: estimated time, maximum depth, machined extent from the origin, cutting length and removed material volume;
- the 2D view of each side with toolpaths, origin and stock dimensions, and two dimetric views of the simulation (side 1 and side 2);
- the tool loading order, then the toolpath table: operation, tool, depth, passes, allowance, entry, tabs, feed, plunge and time;
- the points to watch it detects (cut through the stock, cut-out without tabs, pass too deep, job outside the stock, hidden toolpaths);
- a checklist before starting, a notes area and sign-off boxes.

Save as PDF (in the sheet window, or straight from **File** › **Save the setup sheet as PDF** without opening it) creates an A4 PDF file, split into pages without cutting tables or pictures, ready to print or send. The PDF is a picture of the sheet (text not selectable); for a PDF with text, use **Print** then the browser's “Save as PDF”. You can also keep the HTML file with the G-code. The sheet settings are saved with the project.

Install and use offline

- **Online**: open the GitHub Pages site of the COPO repository. Nothing to install.
- **As an app** (Chrome, Edge, Brave, Vivaldi… on Windows, macOS and Linux): **Help › Install COPO as an app…** or the install icon in the address bar. If the icon does not appear, use the browser menu: in Chrome, **⋮ › Cast, save and share › Install page as app…**; in Edge, **… › Apps › Install this site as an app**. If COPO is already installed, the browser offers “Open in COPO” instead. COPO then opens in its own window with its icon in the app menu, works **offline** once loaded, and **double-clicking a .copo file** opens it directly (the file icon depends on the system). Updates arrive automatically when online. Firefox and Safari do not offer this install: COPO works normally there in a tab.
- **Offline**: download the complete COPO folder (“Code › Download ZIP” on GitHub) and unzip it. Open `index.html` with a double-click: everything is already in the `lib` folder (calculation libraries and interface fonts) and in the `fonts` folder. Nothing else to install or download.
- **Interface fonts**: the `lib/fonts` folder holds the COPO fonts (Source Sans 3, JetBrains Mono, Barlow Condensed). Offline, the interface, the setup sheet and this guide keep the same look.
- **Engraving fonts**: put your `.ttf`, `.otf` or `.woff` files in a `fonts` folder next to COPO.html and open it once from the File menu.

COPO runs on Windows, macOS and Linux, in a recent Chrome, Edge or Firefox. The list of installed fonts is only available in Chrome and Edge.

Keyboard shortcuts

KEY	ACTION	KEY	ACTION
`V`	Select	`Ctrl` + `Z` / `Y`	Undo / Redo
`N`	Nodes	`Ctrl` + `S` / `O`	Save / Open a project
`H` or `Space`	Pan the view	`Ctrl` + `D`	Duplicate
`R` `E` `C`	Rectangle, ellipse, circle	`Ctrl` + `A`	Select all
`O`	Polygon and star	`Ctrl` + `G`	Group (`Shift` : ungroup)
`P` `A` `T`	Pen, arc, text	`Esc` / right-click	Back to selection, deselect
`K` `F`	Chamfer, fillet	`Del`	Delete the selection or node
`X` `U`	Trim, cut	Arrows	Move by 1 mm (`Shift` : 10 mm)
`M` `D`	Measure, dimension	`Esc`	Cancel the current path, deselect

On a Mac, `Cmd` replaces `Ctrl`.

Report a bug

Help › **Report a bug…** menu: describe the problem, then click **Open the report on GitHub**. The report arrives pre-filled with technical information (browser, units, stock, toolpaths, recent errors). The workspace screenshot is copied to the clipboard: paste it into the report with `Ctrl` + `V`. If you tick “Attach the project”, drag the downloaded file into the report. A free GitHub account is enough. Ideas for improvements are welcome too at github.com/gbulleryahen-hash/copo/issues.

Help › **Suggest an improvement…** opens the same window for an idea: describe the feature you want and what it would be used for; it reaches GitHub with the *enhancement* label. Both carry the COPO **version number** (also shown in **Help** › **About**): 1.*n*, where *n* goes up with every published update.

Licence

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Any modified version you distribute must stay under the same licence, with its source code. COPO comes with no warranty.

Libraries used, bundled in the `lib` folder: Clipper (Boost Software License), opentype.js (MIT), d3-delaunay and delaunator (ISC).

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